COMPUTER NETWORKS

BCA

Presented By

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Network

Network is a group of computers connected with each other through wires, optical fibres or optical links so that various devices can interact with each other through a network.

Networking

Computer networking refers to interconnected computing devices that can exchange data and share resources with each other.

Overview of Data Communication Networks

Communication is defined as a process in which more than one computer transfers information, instructions to each other and for sharing resources. Or in other words, communication is a process or act in which we can send or receive data.

The effectiveness of any data communications system depends upon the following **four** fundamental characteristics:

Components of Data Communication

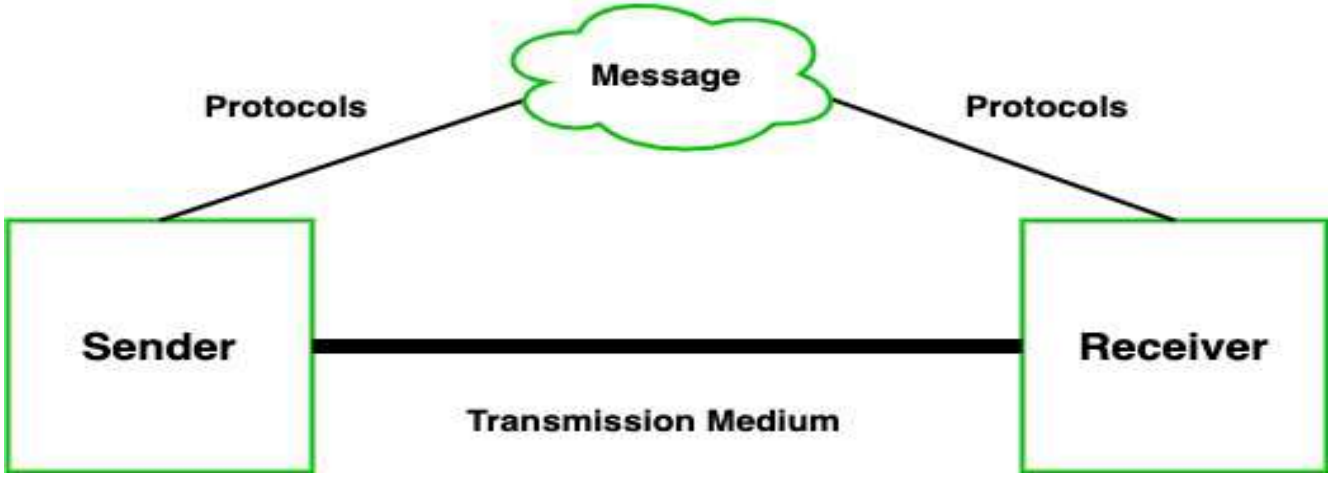
Message: A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.

Sender: It is simply a device that sends data messages. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.

Receiver: It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.

Transmission Medium / Communication Channels: Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless media.

Set of rules (Protocol): When someone sends the data (The sender), it should be understandable to the receiver also otherwise it is meaningless. For example, Sonali sends a message to Chetan. If Sonali writes in Hindi and Chetan cannot understand Hindi, it is a meaningless conversation.



Type of data communication

The data communication is divided into three types:

1. **Simplex Communication**: It is one-way communication or we can say that unidirectional communication in which one device only receives and another device only sends data and devices uses their entire capacity in transmission. For example, IoT, entering data using a keyboard, listing music using a speaker, etc.

2. **Half Duplex communication**: It is a two-way communication, or we can say that it is a bidirectional communication in which both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. For example, walkie-talkie.

3. **Full-duplex communication:** It is a two-way communication or we can say that it is a bidirectional communication in which both the devices can send and receive data at the same time. For example, mobile phones, landlines, etc.

Protocol and Standard in Computer Networks

Protocols and standards are important in computer networks. They are like the rules and guidelines that allow different devices and systems to communicate and work together smoothly. Protocols define how data is sent, received, and processed, while standards ensure that various technologies are compatible with each other.

Network protocol ensures that different technologies and components of the network are compatible with one another, reliable, and able to function together.

What is Protocol?

A protocol is a set of rules that determines how data is sent and received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly. Protocols help make sure that data moves smoothly and securely between devices on a network.

In the above diagrams, Protocols are shown as a set of rules. Communication between the Sender and Receiver is not possible without Protocol.

Key Elements of Protocol

- **Timing:** Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data. Timing ensures preventing of data loss, collisions, and other timing-related issues.

- **Sequence Control**: Sequence control ensures the proper ordering of data packets. The main responsibility of sequence control is to acknowledge the data while it get received, and the retransmission of lost data. Through this mechanism, the data is delivered in correct order.

- **Flow Control:** Flow control regulates device data delivery. It limits the sender's data or asks the receiver if it's ready for more. Flow control prevents data congestion and loss.

- **Error Control:** Error control mechanisms detect and fix data transmission faults. They include error detection codes, data resend, and error recovery. Error control detects and corrects noise, interference, and other problems to maintain data integrity.

- **Security :** Network security protects data confidentiality, integrity, and authenticity. which includes encryption, authentication, access control, and other security procedures. Network communication's privacy and trustworthiness are protected by security standards.

Types of Protocol

Network Layer Protocols

Transport Layer Protocols

Application Layer Protocol

Wireless Protocols

Routing Protocols

Security Protocols

Internet Protocols

Important Protocols Used in Computer Network

TCP (Transmission Control Protocol): Ensures data is sent and received accurately by breaking it into packets, sending them, and reassembling them at the destination.

IP (Internet Protocol): Addresses and routes the packets to make sure they reach the right destination.

HTTP/HTTPS (Hypertext Transfer Protocol/Secure) HTTP used for transferring web pages on the internet.

FTP (File Transfer Protocol) Used for transferring files between computers on a network. It allows users to upload and download files.

SMTP (Simple Mail Transfer Protocol): Used for sending emails. It transfers emails from a client to a server or between servers.

DNS (Domain Name System): **DNS:** DNS stands for Domain Name System. An IP address is used to identify the connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses. Therefore, the system that maps the name to the address is known as Domain Name System.

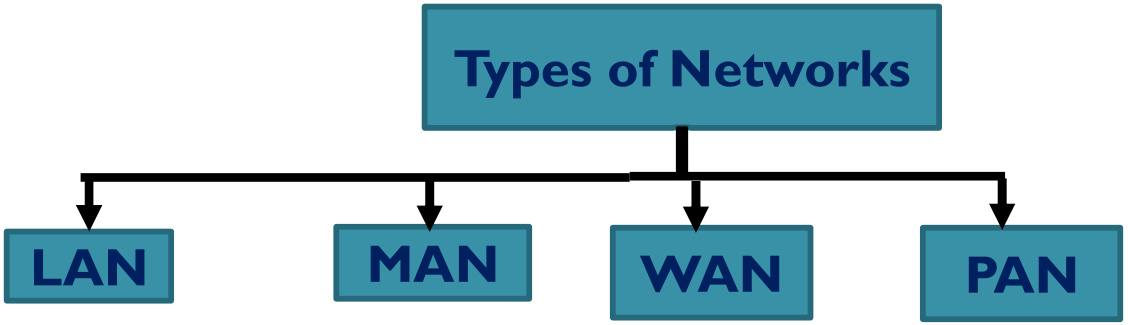
DHCP (Dynamic Host Configuration Protocol): Automatically assigns IP addresses to devices on a network, ensuring each device has a unique address.

SSH (Secure Shell): Provides a secure way to access and manage devices over a network. It encrypts the data, making it safe from eavesdropping.

SNMP (Simple Network Management Protocol): Used for managing and monitoring network devices like routers, switches, and servers. It collects and organizes information about these devices

Types of Networks

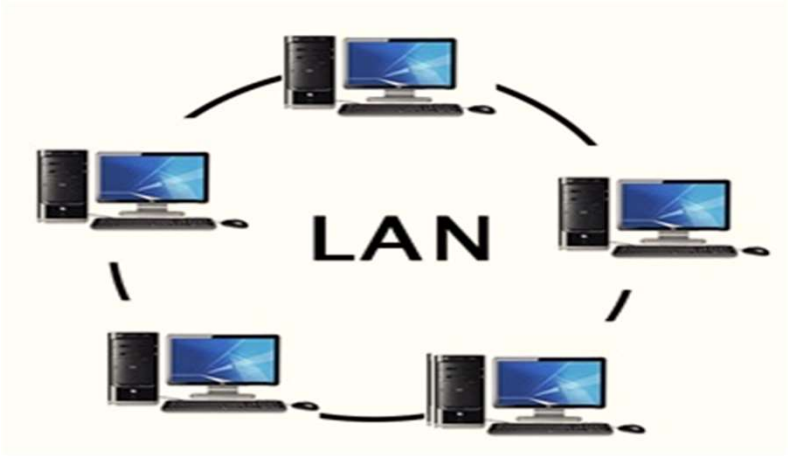
- Local Area Network (LAN)
- Wide Area Network (WAN)
- Metropolitan Area Network (MAN)
- Personal Area Network (PAN)



LAN (Local Area Network)

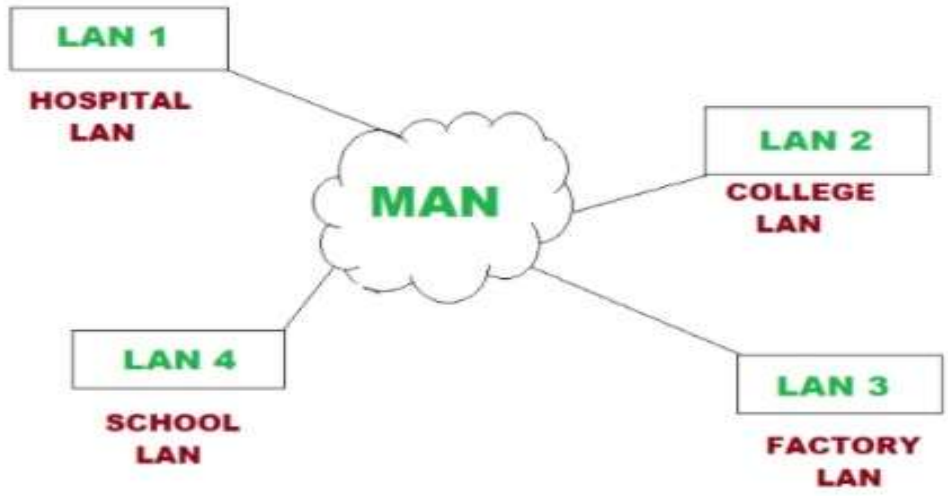
- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and Ethernet cables.

- **The data is transferred at an extremely faster rate in Local Area Network.**



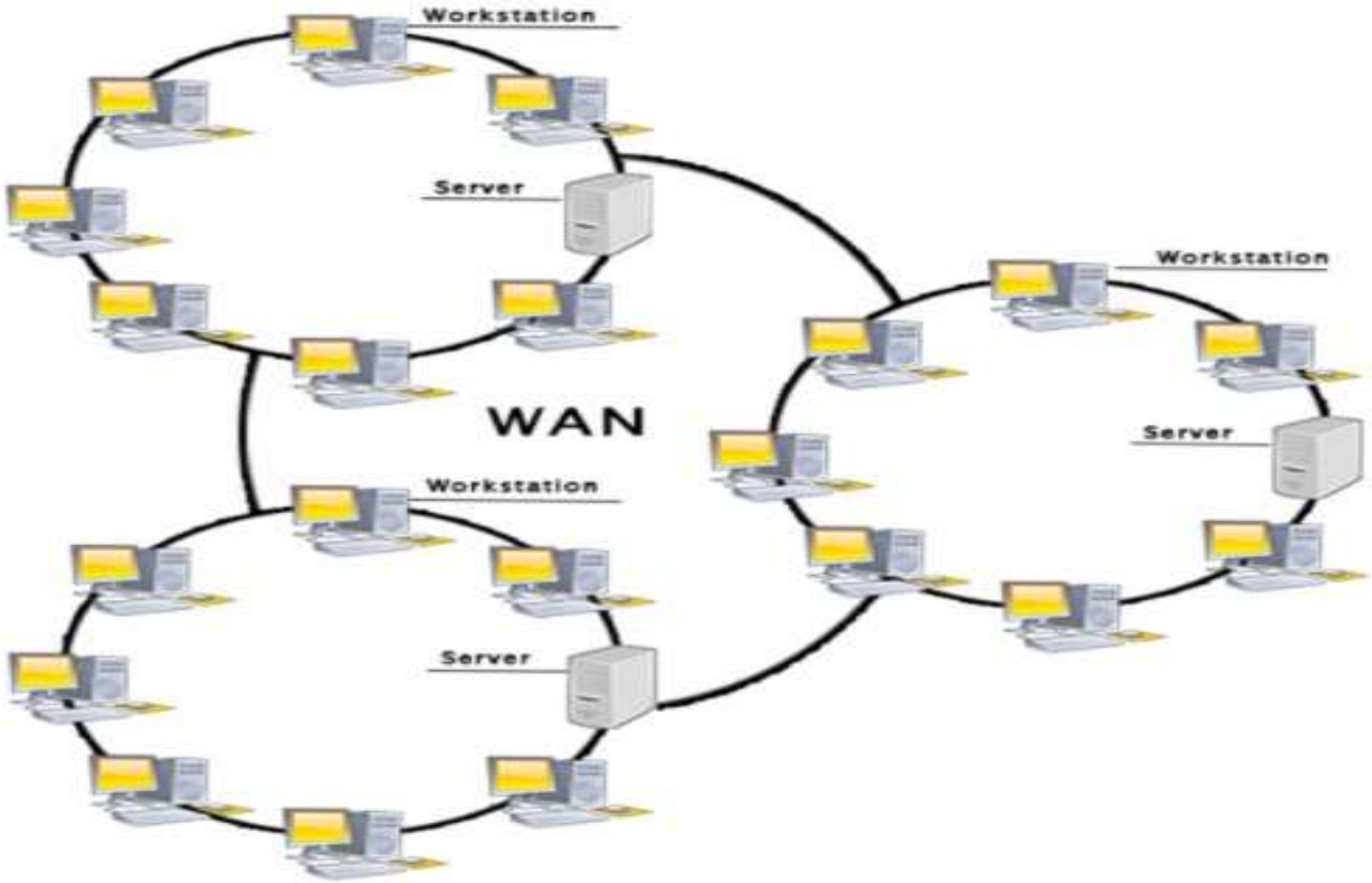
Metropolitan Area Network (MAN)

MAN or Metropolitan area Network covers a larger area than that of a LAN and smaller area as compared to WAN. It connects two or more computers that are apart but resides in the same or different cities. It covers a large geographical area and may serve as an ISP (Internet Service Provider).



WAN(Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN & MAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.


Workstation
Server
Workstation
WAN
Server
Workstation
Server

Examples Of Wide Area Network:

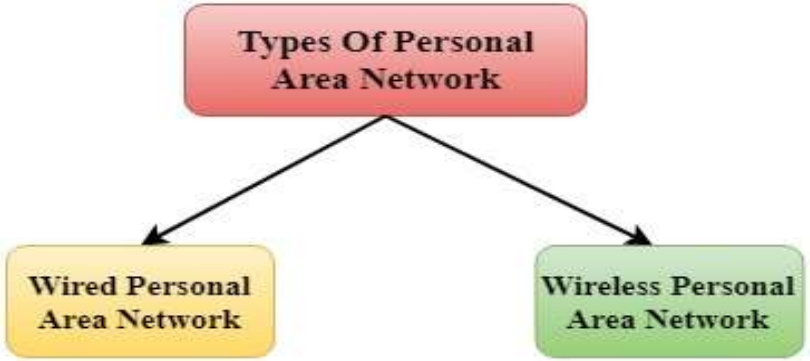
- **Mobile Broadband:** A 4G network is widely used across a region or country.
- **Last mile:** A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.
- **Private network:** A bank provides a private network that connects the 44 offices. This network is made by using the telephone leased line provided by the telecom company.

PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of 30 feet.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.

There are two types of Personal Area Network:

- **Wired Personal Area Network**
- **Wireless Personal Area Network**



Wireless Personal Area Network: Wireless Personal Area Network is developed by simply using wireless technologies such as Wi-Fi, Bluetooth. It is a low range network.

Wired Personal Area Network: Wired Personal Area Network is created by using the USB.

Examples Of Personal Area Network:

- **Body Area Network:** Body Area Network is a network that moves with a person.

For example, *a m*obile network moves with a person. Suppose a person establishes a network connection and then creates a connection with another device to share the information.

- **Offline Network:** An offline network can be created inside the home, so it is also known as a **home network**. A home network is designed to integrate the devices such as printers, computer, television but they are not connected to the internet.
- **Small Home Office:** It is used to connect a variety of devices to the internet and to a corporate network using a *VPN.*

OSI Model & TCP/IP

What Is the OSI Model

The Open Systems Interconnection (OSI) model describes seven layers that computer systems use to communicate over a network. It was the first standard model for network communications, adopted by all major computer and telecommunication companies in the early 1980s

The modern Internet is not based on OSI, but on the simpler TCP/IP model. However, the OSI 7-layer model is still widely used, as it helps visualize and communicate how networks operate, and troubleshoot networking problems.

OSI was introduced in 1983 by representatives of the major computer and telecom companies, and was adopted by ISO as an international standard in 1984.

Introduction OSI

- It divides network architecture into seven layers which, from top to bottom, are the Application, Presentation, Session, Transport, Network, Data Link, and Physical Layers. It is therefore often referred to as the **OSI Seven Layer Model**.

OSI History

- In 1978, the International Standars Organization (ISO) began to develop its OSI framework architecture.

- OSI has two major components: an abstract model of networking, called the Basic Reference Model or seven-layer model, and a set of specific protocols.

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

OSI Layers

OSI Model			
	Data unit	Layer	Function
Host layers	Data	7. Application	Network process to application
		6. Presentation	Data representation, encryption and decryption
		5. Session	Interhost communication
	Segments	4. Transport	End-to-end connections and reliability, Flow control
Media layers	Packet	3. Network	Path determination and logical addressing
	Frame	2. Data Link	Physical addressing
	Bit	1. Physical	Media, signal and binary transmission

Going from layer 1 to 7: **Please Do Not Throw Sausage Pizza Away**
Going from layer 7 to 1: **All People Seem To Need Data Processing**

Application Layer

The application layer is used by end-user software such as web browsers and email clients. It provides protocols that allow software to send and receive information and present meaningful data to users. A few examples of application layer protocols are the **Hypertext Transfer Protocol** (HTTP), File Transfer Protocol (FTP), Post Office Protocol (POP), Simple Mail Transfer Protocol (SMTP), and Domain Name System (DNS).

Presentation Layer

The presentation layer prepares data for the application layer. It defines how two devices should encode, encrypt, and compress data so it is received correctly on the other end. The presentation layer takes any data transmitted by the application layer and prepares it for transmission over the session layer.

Presentation layer protocol include MPEG and HTTP/HTML etc.

Session Layer

The session layer creates communication channels, called sessions, between devices. It is responsible for opening sessions, ensuring they remain open and functional while data is being transferred, and closing them when communication ends. The session layer can also set checkpoints during a data transfer—if the session is interrupted, devices can resume data transfer from the last checkpoint.

Transport Layer

The transport layer takes data transferred in the session layer and breaks it into "segments" on the transmitting end. It is responsible for reassembling the segments on the receiving end, turning it back into data that can be used by the session layer. The transport layer carries out flow control, sending data at a rate that matches the connection speed of the receiving device, and error control, checking if data was received incorrectly and if not, requesting it again.

Network Layer

The network layer has two main functions. One is breaking up segments into network packets, and reassembling the packets on the receiving end. The other is routing packets by discovering the best path across a physical network. The network layer uses network addresses (typically Internet Protocol addresses) to route packets to a destination node.

Data Link Layer

The data link layer establishes and terminates a connection between two physically-connected nodes on a network. It breaks up packets into frames and sends them from source to destination. This layer is composed of two parts—Logical Link Control (LLC), which identifies network protocols, performs error checking and synchronizes frames, and Media **Access Control** (MAC) which uses MAC addresses to connect devices and define permissions to transmit and receive data.

Physical Layer

The physical layer is responsible for the physical cable or wireless connection between network nodes. It defines the connector, the electrical cable or wireless technology connecting the devices, and is responsible for transmission of the raw data, which is simply a series of 0s and 1s, while taking care of bit rate control.

Analog signals were used in many systems to produce signals to carry information. These signals are continuous in both values and time. The use of analog signals has declined with the arrival of digital signals

Examples of Analog Signals

Some common examples of analog signals include:

- Human voice
- Analog Radio and TV Broadcast
- Audio signals transferred via cables
- Radio signals

Digital Signal is a type of communication where the data is communicated in the form of discrete data.
Digital Signal is mostly done in computers, where the data to be transferred is in the form of discrete values. Mostly the data is transferred in the form of binary which has only two discrete values which are zero and one.

Examples of Digital Signals
Some common examples include:

- Digital Audio
- Digital Video
- Binary Data
- Digital Clocks
- Smartphones

DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel.
Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel

Multiplexing

Multiplexing is a technique used to combine and send the multiple data streams over a single medium. The process of combining the data streams is known as multiplexing and hardware used for multiplexing is known as a multiplexer.

Multiplexing is achieved by using a device called Multiplexer (MUX) that combines n input lines to generate a single output line. Multiplexing follows many-to-one, i.e., n input lines and one output line.

Demultiplexing is achieved by using a device called Demultiplexer (DEMUX) available at the receiving end. DEMUX separates a signal into its component signals (one input and n outputs).

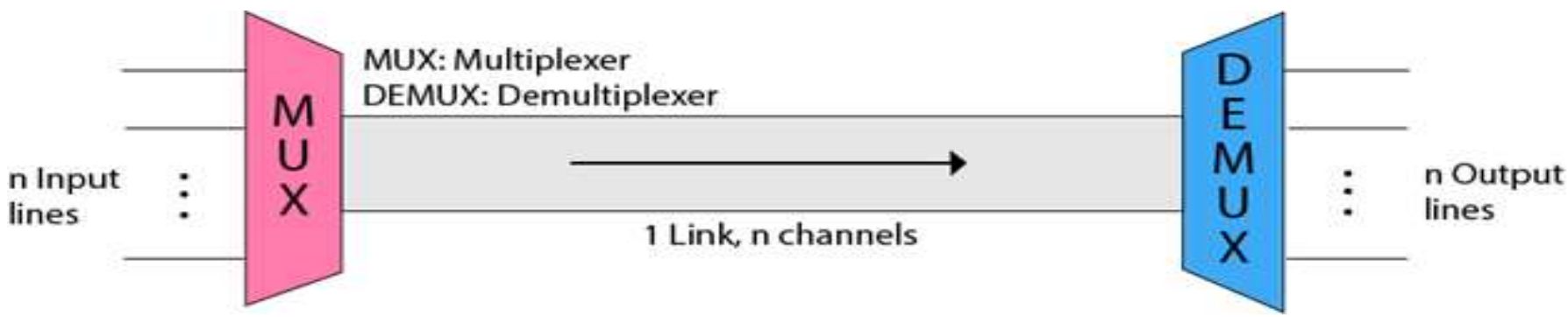
History of Multiplexing

•Multiplexing technique is widely used in telecommunications in which several telephone calls are carried through a single wire.

•Multiplexing originated in telegraphy in the early 1870s and is now widely used in communication.

•George Owen Squier developed the telephone carrier multiplexing in 1910.

Concept of Multiplexing



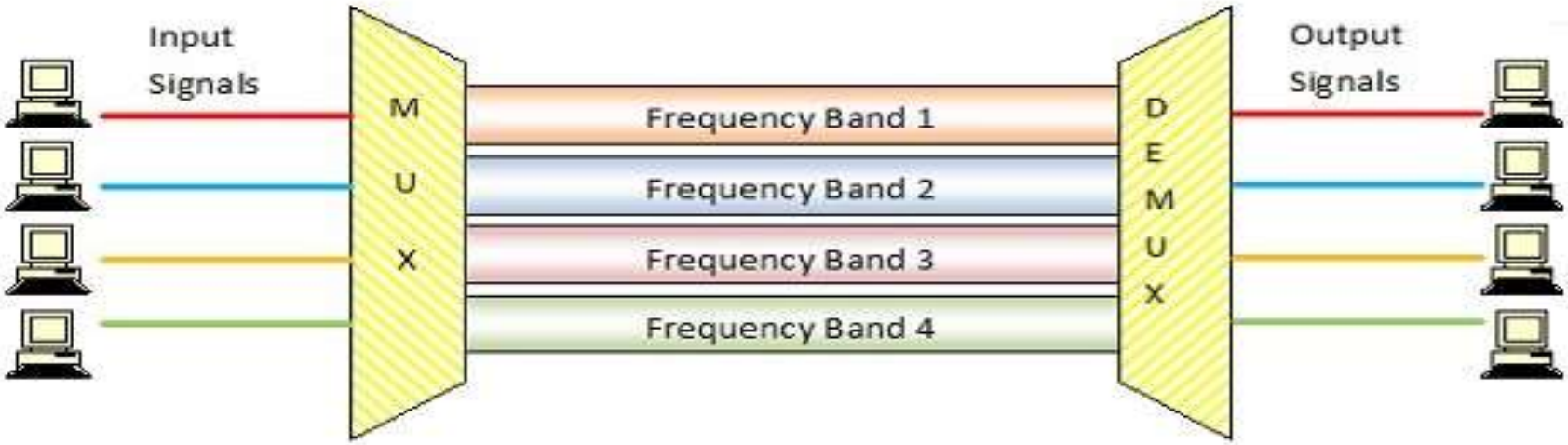
Frequency-division Multiplexing (FDM)

It is an analog technique

Frequency Division Multiplexing is a technique in which the available bandwidth of a single transmission medium is subdivided into several channels.

Frequency division multiplexing (FDM) is a technique of multiplexing which means combining more than one signal over a shared medium.

The modulated signals are combined together using a multiplexer (MUX) in the sending end. The combined signal is transmitted over the communication channel, thus allowing multiple independent data streams to be transmitted simultaneously. At the receiving end, the individual signals are extracted from the combined signal by the process of demultiplexing (DEMUX).



The following diagram conceptually represents multiplexing using FDM. It has 4 frequency bands, each of which can carry signal from 1 sender to 1 receiver. Each of the 4 senders is allocated a frequency band. The four frequency bands are multiplexed and sent via the communication channel. At the receiving end, a demultiplexer regenerates the original four signals as outputs.

Wavelength Division Multiplexing (WDM) is a fiber-optic transmission technique that enables the use of multiple light wavelengths (or colors) to send data over the same medium. Two or more colors of light can travel on one fiber, and several signals can be transmitted in an optical waveguide at differing wavelengths or frequencies on the optical spectrum.

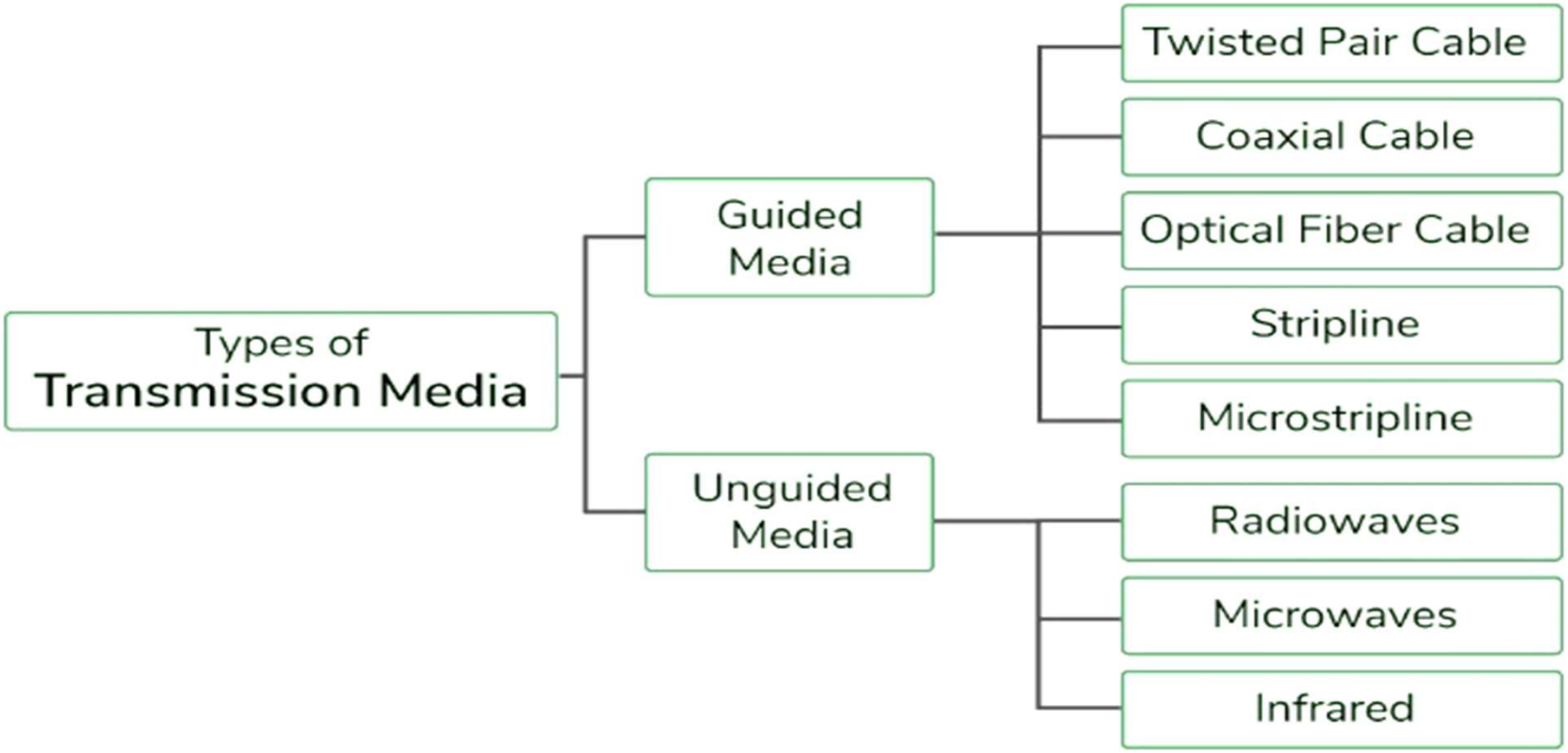
The Time Division Multiplexing (TDM) is a digital procedure. Here, each sender is given the entire possession of the whole bandwidth of the channel for a fixed duration of time. After this, the control is moved to the next sender, and the process continues on a round-robin basis.

An example of TDM is the television broadcast. In a television serial, generally, a 10 minutes' serial is followed by a 5 minutes' advertisement. The time in which the serial is being broadcasted, the total frequency is dedicated to the serial.

Transmission media refer to the physical pathways through which data is transmitted from one device to another within a network. These pathways can be wired or wireless. The choice of medium depends on factors like distance and speed etc.

OR

A Transmission medium is a physical path between the transmitter and the receiver i.e. it is the channel through which data is sent from one place to another. Transmission Media is broadly classified into the following types:



Guided Media

Guided Media is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.

Features:

•High Speed

•Secure

•Used for comparatively shorter distances

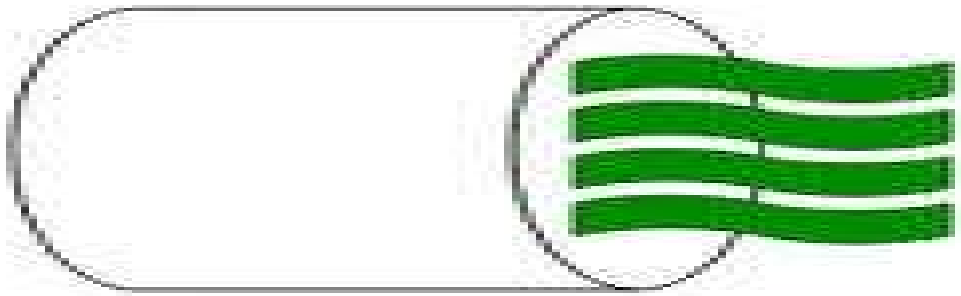
There are 3 major types of Guided Media:

Twisted Pair Cable

It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media.

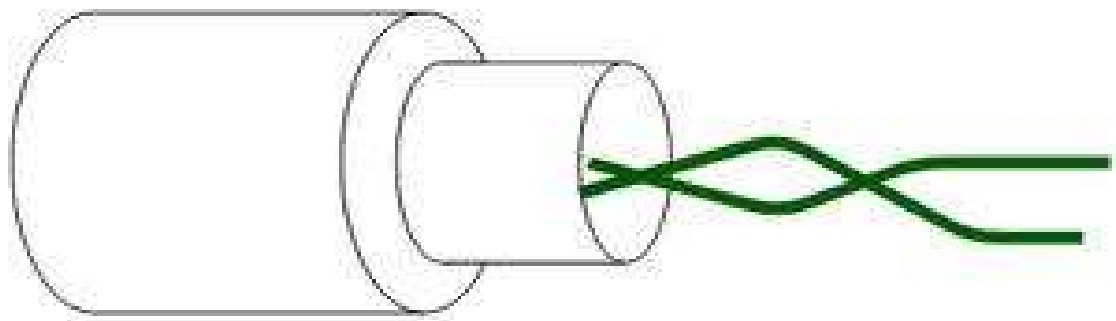
Twisted Pair is of two types:

- **Unshielded Twisted Pair (UTP):** UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.



Unshielded Twisted Pair

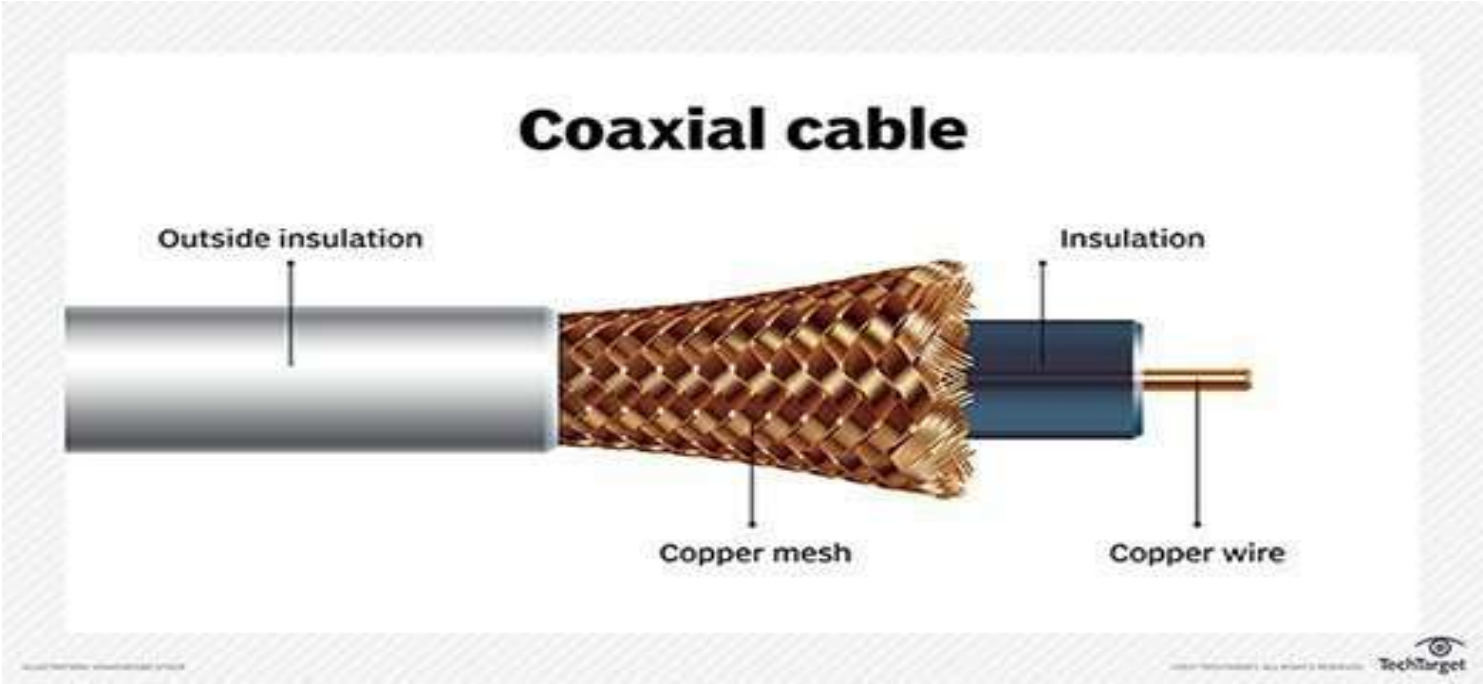
Shielded Twisted Pair (STP): This type of cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.



Shielded Twisted Pair

Coaxial cable is commonly used by cable operators, telephone companies, and internet providers worldwide to convey data, video, and voice communications to customers. It has also been used extensively within homes.

It has an inner conductor surrounded by an insulating layer, surrounded by conductive shielding (outer conductor), and a protective outer jacket.



A fiber optic cable is a network cable that contains strands of glass fibers inside an insulated casing. They're designed for long-distance, high-performance data networking, and telecommunications. Compared to wired cables, fiber optic cables provide higher bandwidth and transmit data over longer distances. Fiber optic cables support much of the world's internet, cable television, and telephone systems.

Unguided Media

An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore it is also known as **wireless transmission**.

In unguided media, air is the media through which the electromagnetic energy can flow easily.

Radio waves are a type of electromagnetic radiation. A radio wave has a much longer wavelength than visible light. Humans use radio waves extensively for communications. This radio tower has both rectangular and circular antennas to transmit and receive radio frequency energy.

Infrared (IR) is a wireless technology that uses light waves to transmit data over short to medium ranges in computer networks. It's similar to the technology used in many household devices, like remote controls and wireless headphones.

S.No.	Guided Media	Unguided Media
1.	In guided media, the signal energy communicates via wires.	In unguided media, the signal energy communicates through the air.
2.	Guided media is generally preferred when we want to execute direct communication.	Unguided media is generally preferred for radio broadcasting in all directions.
3.	The guided media formed the different network topologies.	The unguided media formed the continuous network topologies.
4.	Here, the signals are in the state of current and voltage.	Here, the signals are in the state of electromagnetic waves.
5.	In the case of guided media, the transmission capacity can be boosted by counting more wires.	In the case of unguided media, it is not feasible to acquire more capacity.
6.	Open Wire, Twisted Pair, Coaxial Cable, and Optical Fibre are the different kinds of guided media.	Microwave Transmission, Radio Transmission, and Infrared Transmission are the types of unguided media.

Switching is process to forward packets coming in from one port to a port leading towards the destination. When data comes on a port it is called ingress, and when data leaves a port or goes out it is called egress. A communication system may include number of switches and nodes.

A telephone network is a telecommunications network that connects telephones which allows telephone calls between two or more parties, as well as newer features such as fax and internet.

There are a number of different types of telephone network:

A landline network where the telephones must be directly wired into a single telephone exchange This is known as the **public switched telephone network** or **PSTN.**

A wireless network where the telephones are mobile and can move around anywhere within the coverage area.

A private network where a closed group of telephones are connected primarily to each other and use a gateway to reach the outside world. This is usually used inside companies and call centers and is called a **private branch exchange (PBX).**

Digital Subscriber Lines (DSL)

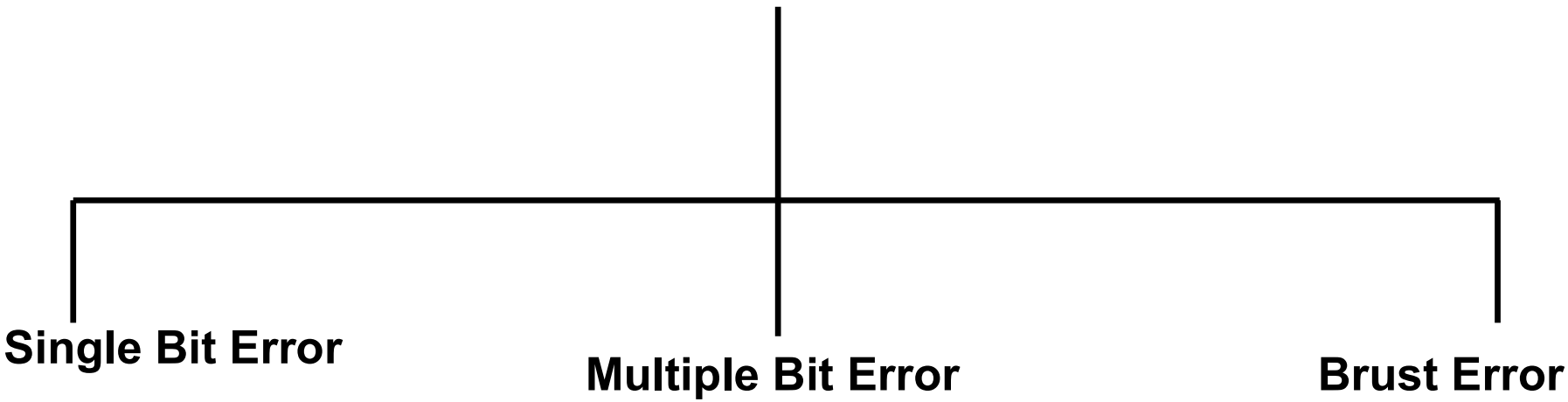
Digital Subscriber Lines (DSL) are a set of technologies that use existing telephone lines to transmit digital data, such as internet access.

DSL uses a modem to convert digital data from a computer into a format that can be transmitted over the telephone line

Error is a condition when the receiver's information does not match the sender's. Digital signals suffer from noise during transmission that can introduce errors in the binary bits traveling from sender to receiver. That means a 0 bit may change to 1 or a 1 bit may change to 0.

Error detection and correction are implemented either at the data link layer or the transport layer of the OSI model.

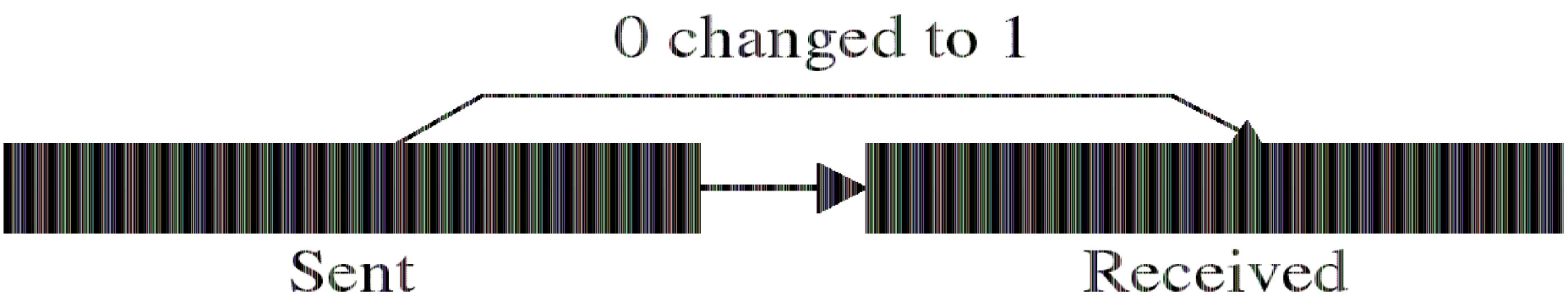
Types of Errors



Single-Bit Error

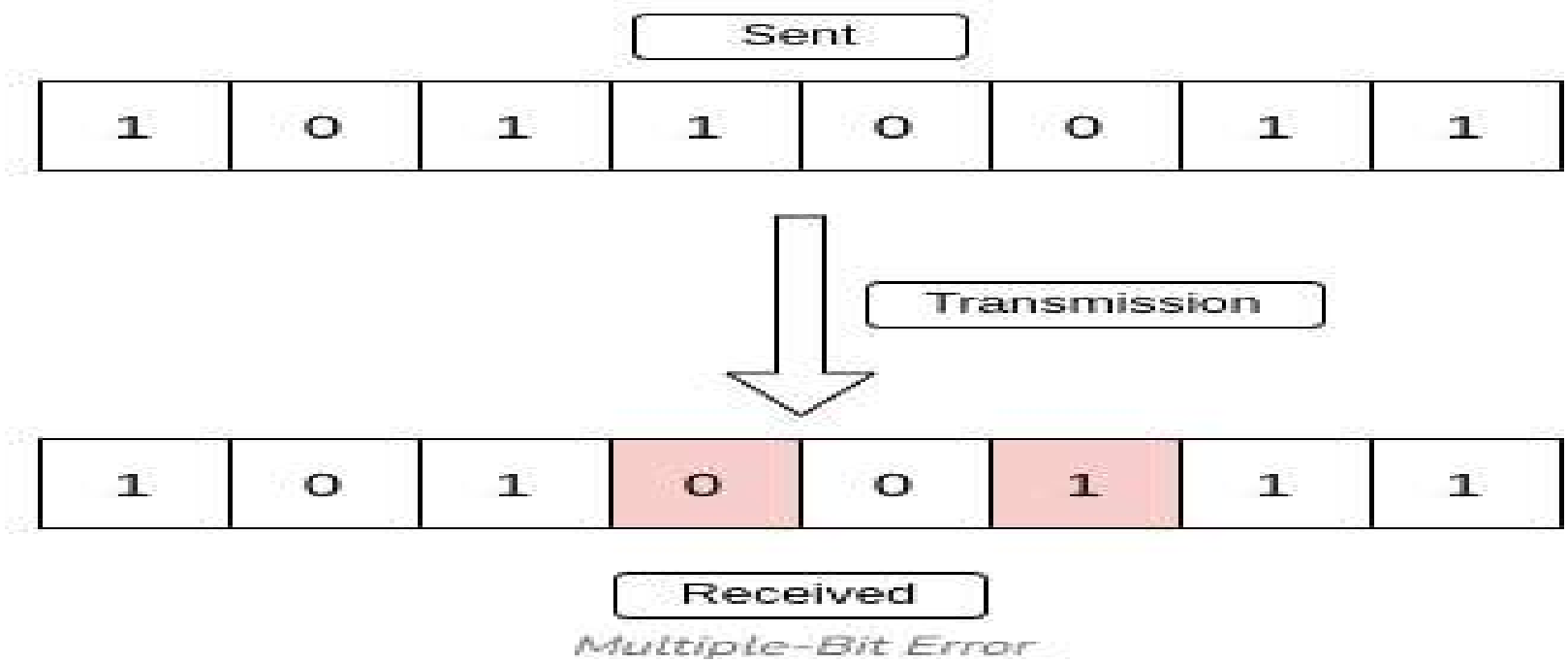
A single-bit error refers to a type of data transmission error that occurs when one bit (i.e., a single binary digit) of a transmitted data unit is altered during transmission, resulting in an incorrect or corrupted data unit.

Single-bit error



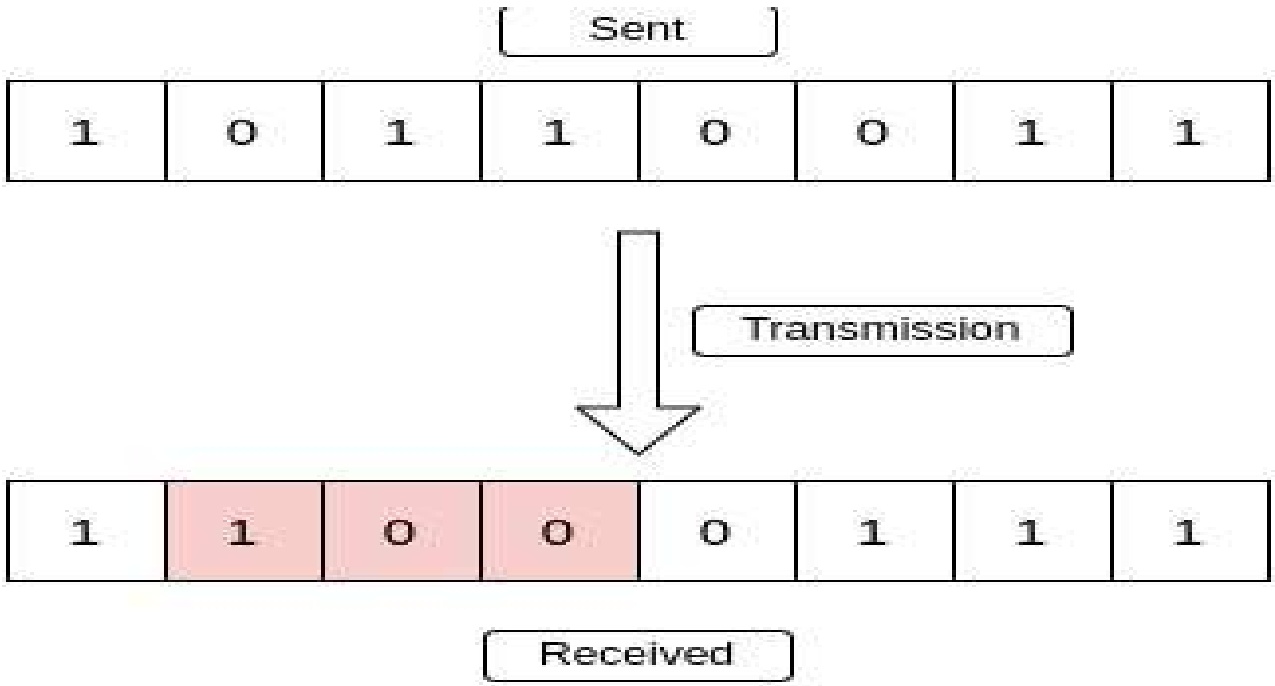
Multiple-Bit Error

A multiple-bit error is an error type that arises when more than one bit in a data transmission is affected. Although multiple-bit errors are relatively rare when compared to single-bit errors, they can still occur, particularly in high-noise or high-interference digital environments.



Burst Error

When several consecutive bits are flipped mistakenly in digital transmission, it creates a burst error. This error causes a sequence of consecutive incorrect values.



Error detection mechanisms are crucial for identifying errors in data during transmission over networks. Here's an overview of common error detection mechanisms, including linear codes, Cyclic Redundancy Check (CRC), and checksum.

1. Linear Codes

•Concept: Linear codes are a type of error-detecting code that use algebraic structures. The simplest form is a parity bit, which ensures that the number of 1s in the data is even (even parity) or odd (odd parity).

•Mechanism: The sender adds an extra bit (parity bit) to each data frame. The receiver checks if the total number of 1s is even or odd, depending on the agreed parity.

•Application: Parity checks are effective at detecting single-bit errors but are limited because they cannot detect multiple-bit errors.

•Examples: Simple parity check, two-dimensional parity, Hamming code (used for error detection and correction).

Cyclic Redundancy Check (CRC)

•Concept: CRC is a widely used error-detection method that involves treating the data as a binary number, dividing it by a fixed binary polynomial, and appending the remainder to the data.

•Mechanism:

- ❖ Both sender and receiver agree on a fixed binary divisor (the generator polynomial).
- ❖ The sender divides the data by this polynomial and appends the remainder (CRC value) to the data.
- ❖ The receiver performs the same division on the received data. If the remainder is zero, the data is assumed to be error-free; otherwise, an error is detected.

•Application: CRC is highly effective at detecting burst errors and is commonly used in network protocols like Ethernet and USB.

•Examples: CRC-8, CRC-16, CRC-32 (where the number indicates the length of the CRC code in bits).

Checksum

•Concept: The checksum is a simple method where data is divided into equal-sized blocks, summed, and the sum's complement is added to the data as the checksum.

•Mechanism:

- Data is divided into fixed-size segments (e.g., 16 bits).
- The segments are added up to produce a sum.
- The sum is complemented (inverted) and appended to the data as the checksum.
- The receiver calculates the checksum by summing the received segments, including the checksum. If the result is zero, the data is assumed to be error-free.

•Application: Checksum is simpler but less robust than CRC and is used in error detection for protocols like TCP/IP, where simpler error-checking is needed.

•Examples: Internet Checksum (used in TCP/IP and UDP protocols).

Error correction mechanisms not only detect errors in transmitted data but also allow the receiver to reconstruct the original data without needing retransmission. These mechanisms are particularly important in environments where retransmissions are costly or impractical, such as satellite communications, deep-space missions, or streaming applications.

Common Error Correction Mechanisms.

1. Hamming Code

•Concept: Hamming code is an error-correcting code that can detect up to two-bit errors and correct single-bit errors in data.

•Mechanism:

- Redundant bits are added to the data in specific positions.
- These bits are calculated based on parity for groups of positions, with each redundant bit covering certain data bits.
- At the receiver's end, the parity bits are recalculated. If there's a discrepancy, the pattern of discrepancies points to the exact bit that is in error, allowing single-bit error correction.

•Application: Used in memory systems, telecommunications, and data storage systems where low levels of errors are anticipated.

•Limitations: Hamming codes are limited to single-bit error correction and are not ideal for handling high error rates.

Data Link Control is a set of protocols and mechanisms used to ensure reliable communication over a network at the Data Link layer (Layer 2) of the OSI model. It includes both flow control and error control, which are essential for handling data frames effectively and preventing issues like data loss, congestion, or duplication.

Flow Control

Flow control is used to manage the rate at which data is transmitted between a sender and a receiver to prevent the receiver from being overwhelmed by too much data too quickly. The two main types of flow control techniques are:

1.Stop-and-Wait:

1. In the Stop-and-Wait protocol, the sender transmits a single frame and then waits for an acknowledgment (ACK) from the receiver before sending the next frame.
2. This simple approach ensures that data is sent in an orderly fashion and prevents the receiver's buffer from overflowing.

2.Sliding Window:

1. In the Sliding Window protocol, the sender can transmit multiple frames before needing an acknowledgment.
2. It uses a "window" of sequence numbers to manage frames that can be sent or received at a time, providing greater efficiency by allowing a continuous flow of frames.
3. This approach is widely used in reliable transmission protocols (like TCP).

Error Control

Error control is responsible for detecting and correcting errors that occur during transmission. It ensures data integrity and can request retransmission if errors are detected. There are three main error control methods:

1.Automatic Repeat request (ARQ):

1. ARQ is a family of protocols that include error detection and retransmission strategies.
2. When an error is detected, a frame is either retransmitted or dropped based on certain conditions, and mechanisms like acknowledgments (ACKs) or negative acknowledgments (NAKs) are used.

2.Forward Error Correction (FEC):

1. FEC adds redundant information to each frame, which allows the receiver to detect and correct errors without needing retransmission.
2. This technique is often used in real-time applications where retransmission would cause unacceptable delays.

3.Cyclic Redundancy Check (CRC):

1. CRC is an error-detection method that attaches a checksum to each frame, allowing the receiver to detect any data corruption that might have occurred during transmission.

Stop-and-Wait ARQ Protocol

The Stop-and-Wait ARQ is a specific flow and error control protocol based on the Stop-and-Wait approach. Here's how it works:

1.Frame Transmission:

1. The sender sends a frame and waits for an acknowledgment from the receiver.
2. If the receiver successfully receives the frame without errors, it sends an acknowledgment (ACK) back to the sender.

2.Acknowledgment and Retransmission:

1. If the sender receives an ACK, it proceeds to send the next frame.
2. If no acknowledgment is received within a certain time frame (due to a frame loss or error), the sender will retransmit the frame.
3. This prevents data loss in case of transmission errors.

3.Error Handling with Timeout:

1. A timer is used to monitor the time taken to receive an ACK. If the timer expires, the sender retransmits the frame, assuming it was lost or corrupted during transmission.

Multiple Access Techniques

Multiple Access techniques determine how multiple devices share the same communication medium efficiently. They can be grouped into three main categories:

1. Random Access

In Random Access, devices access the channel whenever they have data to send, leading to possible collisions, which are then managed by various protocols:

•ALOHA:

- A simple protocol where devices transmit data whenever they are ready.
- Collisions occur when two or more devices send data at the same time. After a collision, devices wait a random time before retrying.
- Variants: Pure ALOHA and Slotted ALOHA, where Slotted ALOHA reduces collisions by allowing transmissions only at specific time slots.

•CSMA (Carrier Sense Multiple Access):

- Devices sense the channel to check if it is idle before transmitting. If busy, they wait until it is free.
- CSMA/CD (Collision Detection): Used in wired networks, especially Ethernet, where devices detect collisions and stop transmission, wait, and then retry.
- CSMA/CA (Collision Avoidance): Used in wireless networks, where collision detection is challenging. Devices wait before transmitting and use acknowledgment signals to minimize collisions.

Controlled Access

In Controlled Access, a centralized control mechanism or protocol determines which device can access the medium at any time, which helps avoid collisions:

•Polling:

- A controller (like a primary device or access point) polls each device in turn, asking if it has data to transmit.
- Effective in environments where control and reliability are priorities, such as in point-to-multipoint communication.

•Reservation:

- Devices reserve the channel for specific time slots or channels, typically in advance, reducing the likelihood of collision.
- Often used in time-sensitive applications where reliable access is required, like satellite and cellular networks.

•Token Passing:

- A "token" (a control frame) circulates among devices, and only the device holding the token can transmit data.
- Token Ring and Token Bus networks use this method, effectively eliminating collisions but requiring coordination.

Channelization

Channelization divides the channel into distinct sections that multiple users can access simultaneously, usually by frequency, time, or code:

•FDMA (Frequency Division Multiple Access):

- The available bandwidth is divided into frequency bands, with each device assigned its own band.
- Used in radio and satellite communications.

•TDMA (Time Division Multiple Access):

- Devices take turns transmitting data in designated time slots.
- Widely used in cellular networks, allowing multiple users on the same frequency.

•CDMA (Code Division Multiple Access):

- Devices use unique codes to encode their transmissions, allowing multiple transmissions on the same frequency.
- Commonly used in cellular networks, where multiple users can share the same spectrum without interfering with each other.

Wired LANs (Ethernet)

Ethernet is the dominant standard for wired LANs, known for its simplicity, scalability, and cost-effectiveness. It has evolved over the years, resulting in different types of Ethernet standards:

1. Traditional Ethernet (10 Mbps)

•Original Ethernet, based on the IEEE 802.3 standard, operates at 10 Mbps over coaxial cable or twisted-pair cables.

•Uses CSMA/CD for medium access control in half-duplex mode (where only one device can transmit at a time).

2. Fast Ethernet (100 Mbps)

•Fast Ethernet (IEEE 802.3u) increased the speed to 100 Mbps.

•Uses twisted-pair (Category 5) or fiber-optic cables.

•Supports both half-duplex (CSMA/CD) and full-duplex modes, allowing simultaneous sending and receiving without collisions.

•Common variations: 100BASE-TX (twisted-pair) and 100BASE-FX (fiber-optic).

3. Gigabit Ethernet (1 Gbps)

•Gigabit Ethernet (IEEE 802.3z for fiber and IEEE 802.3ab for twisted pair) offers data rates of 1 Gbps.

•Uses Category 5e or higher for twisted-pair or fiber-optic cables.

•Typically operates in full-duplex mode, eliminating the need for CSMA/CD due to dedicated send/receive channels.

•Common variations include 1000BASE-T (twisted-pair) and 1000BASE-LX (fiber-optic).

10 Gigabit Ethernet (10 Gbps and Beyond)

•Newer standards like 10 Gigabit Ethernet (IEEE 802.3ae) and beyond (40 Gbps, 100 Gbps) have been introduced for very high-speed networks.

•Primarily used in data centers and high-performance applications.

•Operates exclusively in full-duplex mode, typically using fiber-optic cables, though there are copper-based variants (like 10GBASE-T).

Summary of Ethernet Standards

Ethernet Standard	Speed	Cable Type	Duplex Modes	Use Cases
Traditional	10 Mbps	Coaxial, Twisted Pair	Half and Full	Small, early LANs
Fast Ethernet	100 Mbps	Twisted Pair, Fiber	Half and Full	Medium-sized LANs
Gigabit Ethernet	1 Gbps	Twisted Pair, Fiber	Full	Larger LANs, offices
10 Gigabit+	10 Gbps+	Fiber, High-Grade Twisted Pair	Full	Data centers, high-performance

M-III

Network Devices: Network devices, also known as networking hardware, are physical devices that allow hardware on a computer network to communicate and interact with one another. For example Repeater, Hub, Bridge, Switch, Routers, Gateway, and NIC, etc.

Repeater – A repeater operates at the physical layer. Its job is to amplifies (i.e., regenerates) the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-port device.

Hub – A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices.

Types of Hub

- **Active Hub**:- These are the hubs that have their power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.

- **Passive Hub**:- These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.

- **Intelligent Hub**:- It works like an active hub and includes remote management capabilities. They also provide flexible data rates to network devices. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

A layer 2 switch is a type of network switch or device that works on the data link layer (OSI Layer 2) and utilizes MAC Address to determine the path through where the frames are to be forwarded. It uses hardware based switching techniques to connect and transmit data in a local area network (LAN).

•MAC Address Table: A Layer-2 switch will keep a MAC address table that identifies the MAC address of the connected device and the port number. This enables the switch to direct frames towards the correct destination.

•VLAN Support: Layer-2 switches can support VLANs (Virtual Local Area Networks), which means that the network can be segmented at this layer.

•Switching Frames: These forward Ethernet frames are according to MAC addresses and are vital in minimizing collision within LANs.

•Broadcast Domains: Layer-2 switches do not partition broadcast domains, all the devices connected to a Layer-2 switch are in the same broadcast domain.

ROUTER

A router is a device that connects two or more packet-switched networks or subnetworks. It serves two primary functions: managing traffic between these networks by forwarding data packets to their intended IP addresses, and allowing multiple devices to use the same Internet connection.

There are several types of routers, but most routers pass data between LANs (local area networks) and WANs (wide area networks). A LAN is a group of connected devices restricted to a specific geographic area. A LAN usually requires a single router.

A WAN, by contrast, is a large network spread out over a vast geographic area. Large organizations and companies that operate in multiple locations across the country, for instance, will need separate LANs for each location, which then connect to the other LANs to form a WAN. Because a WAN is distributed over a large area, it often necessitates multiple routers and switches.

What is a Layer 3 switch?

Also called a multilayer switch, it is a specialized hardware device that has a lot in common with the traditional router—both in physical appearance and function. Layer 3 switches support the same routing protocols as routers and inspect incoming packets, as well as make vital routing decisions the same way routers do. And they do these routing tasks in addition to performing switching duties. Like routers, Layer 3 switches can be configured to support such routing protocols as:

- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Enhanced Interior Gateway Routing Protocol (EIGRP)

What is the purpose of Layer 3 switches?

These switches were designed to improve the network routing performance on large local area networks (LANs) such as corporate intranets.

While Layer 2 switches can route data traffic dynamically when there is low to medium traffic, their performance suffers when it increases to a high level.

The main difference between a router and a Layer 3 switch lies inside the hardware itself. Layer 3 switch hardware combines that of traditional switch and routers—replacing some of the typical router software with integrated circuit hardware that provides better LAN performance. In addition, a Layer 3 switch will not usually have wide area network (WAN) ports which are standard in routers.

What are the main features of a Layer 3 switch?

These switches feature the following:

- Performance on two OSI layers: Layer 2 and Layer3
- Usually come in 24 or 48 Ethernet
- Uses a simple switching algorithm
- Routing protocols are simple

Gateway

A gateway is a network node or device that connects two networks that use different transmission protocols. Gateways play an important role in connecting two networks. It works as the entry-exit point for a network because all traffic that passes across the networks must pass through the gateway.
The gateway monitors and controls all the incoming and outgoing network traffic.

IPV4 AND IPV6

What is IP?

An IP, or Internet Protocol address, is a unique set of numbers assigned to each device connected to a network, like the Internet. It's like an address for your computer, phone, or any other device, allowing them to communicate with each other. When you visit a website, your device uses the IP address to find and connect to the website's server.

Difference Between IPv4 and IPv6

The address through which any computer communicates with our computer is simply called an Internet Protocol Address or IP address. For example, if we want to load a web page or download something, we require the address to deliver that particular file or webpage. That address is called an IP Address.

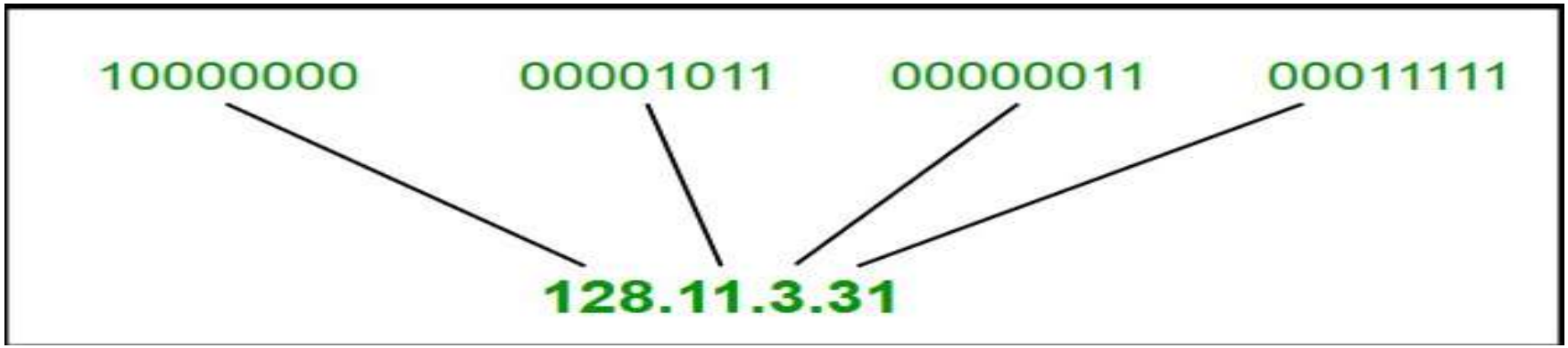
There are two versions of IP: IPv4 and IPv6. IPv4 is the older version, while IPv6 is the newer one. Both have their own features and functions

IP stands for Internet Protocol version v4 stands for Version Four (IPv4), is the most widely used system for identifying devices on a network. It uses a set of four numbers, separated by periods (like 192.168.0.1), to give each device a unique address. This address helps data find its way from one device to another over the internet.

IPv4 addresses are 32-bit integers that have to be expressed in Decimal Notation. It is represented by 4 numbers separated by dots in the range of 0-255, which have to be converted to 0 and 1, to be understood by Computers. For Example, An IPv4 Address can be written as 192.168.0.50.

IPv4 Address Format

IPv4 Address Format is a 32-bit Address that comprises binary digits separated by a dot (.).



IPV4 Format

What is IPv6?

IPv6 is based on IPv4 and stands for Internet Protocol version 6. It was first introduced in December 1995 by Internet Engineering Task Force. IP version 6 is the new version of Internet Protocol, which is way better than IP version 4 in terms of complexity and efficiency. IPv6 is written as a group of 8 hexadecimal numbers separated by colon (:). It can be written as 128 bits of 0s and 1s.

IPv6 Address Format

IPv6 Address Format is a 128-bit IP Address, which is written in a group of 8 hexadecimal numbers separated by colon (:).

ABCD:EF01:2345:6789:ABCD:B201:5482:D023

IP stands for Internet Protocol and v4 stands for Version Four (IPv4). IPv4 was the primary version brought into action for production within the ARPANET in 1983. IP version four addresses are 32-bit integers which will be expressed in decimal notation. In this article, we will discuss about IPv4 datagram header.

What is IPv4?

IPv4 is a connectionless protocol used for packet-switched networks. Internet Protocol Version 4 (IPv4) is the fourth revision of the Internet Protocol and a widely used protocol in data communication over different kinds of networks. IPv4 is a connectionless protocol used in packet-switched layer networks, such as Ethernet. It provides a logical connection between network devices by providing identification for each device.

There are many ways to configure IPv4 with all kinds of devices – including manual and automatic configurations – depending on the network type. IPv4 uses 32-bit addresses for Ethernet communication in five classes: A, B, C, D and E. Classes A, B, and C have a different bit length for addressing the network host. Class D addresses are reserved for multicasting, while class E addresses are reserved for military purposes.

IPv4 Datagram Header

•**VERSION**: Version of the IP protocol (4 bits), which is 4 for IPv4

•**HLEN**: IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.

•**Type of service:** Low Delay, High Throughput, Reliability (8 bits)

•**Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.

•**Identification**: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)

•**Flags**: 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)

•**Fragment Offset**: Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.

•**Time to live**: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.

•**Protocol**: Name of the protocol to which the data is to be passed (8 bits)

•**Header Checksum**: 16 bits header checksum for checking errors in the datagram header

•**Source IP address**: 32 bits IP address of the sender

•**Destination IP address**: 32 bits IP address of the receiver

IPv6 packet format has several advantages, including:

- **Larger address space**

•IPv6 packets have a 128-bit address, which is much larger than IPv4's 32-bit address. This means there are almost an unlimited number of unique IP addresses, which is important for the internet's growth.

- **Streamlined header**

•IPv6 packets have a fixed-length header of 40 bytes, which is faster to process than IPv4's header.

- **Flow labeling**

•IPv6 packets can be labeled to indicate special handling, such as real-time or non-default quality of service.

- **Jumbograms**

•IPv6 packets can handle payloads larger than 2^32, which improves performance on high maximum transmission unit (MTU) links.

- **Fragmentation and reassembly**

•IPv6 packets are fragmented and reassembled by the sender and receiver, not routers, which makes routing more efficient.

- **Multicast**

•IPv6 uses multicast to send a single packet to multiple destinations at once, which saves bandwidth and node processing resources.

- **IPsec integration**

•IPv6 includes IPsec encryption and authentication to protect against attacks and ensure data privacy.

- **Automatic address configuration**

•IPv6 includes features like link-local addressing to simplify the process of assigning IP addresses to devices.

•No network address translation (NAT)

•IPv6 eliminates the need for NAT, which reduces complexity and processing overhead.

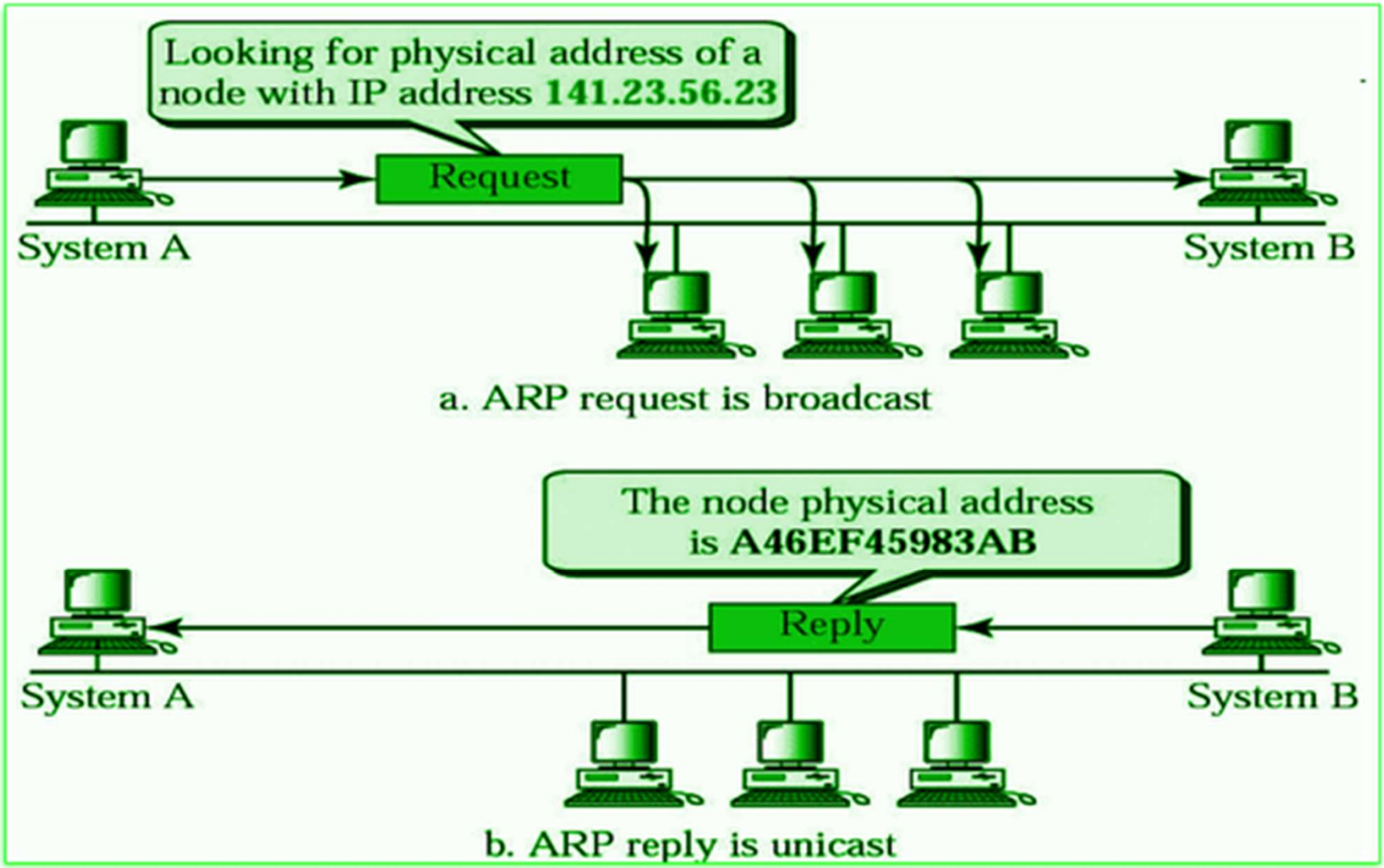
MODULE-IV

Network Layer Protocols

Network Layer is responsible for the transmission of data or communication from one host to another host connected in a network. Rather than describing how data is transferred, it implements the technique for efficient transmission. In order to provide efficient communication protocols are used at the network layer. The data is being grouped into packets or in the case of extremely large data it is divided into smaller sub packets. Each protocol used has specific features and advantages. The below article covers in detail the protocols used at the network layer.

ARP (Address Resolution Protocol)

ARP stands for Address Resolution Protocol. ARP is used to convert the logical address ie. IP address into physical address ie. MAC address. While communicating with other nodes, it is necessary to know the MAC address or physical address of the destination node. If any of the node in a network wants to know the physical address of another node in the same network, the host then sends an ARP query packet. This ARP query packet consists of IP address and MAC address of source host and only the IP address of destination host. This ARP packet is then received to every node present in the network. The node with its own IP address recognises it and sends it MAC address to the requesting node. But sending and receiving such packets to know the MAC address of destination node it increases the traffic load. Therefore in order to reduce this traffic and improve the performance, the systems that makes use of ARP maintain a cache of recently acquired IP into MAC address bindings.



Types of ARP Entries

•**Static Entry:** This type of entry is created when a user uses the ARP command utility to manually enter the IP to MAC address association.

•**Dynamic Entry:** A dynamic entry is one that is automatically formed when a sender broadcasts their message to the whole network. Dynamic entries are periodically removed and are not permanent.

RARP(Reverse Address Resolution Protocol)

RARP stands for Reverse Address Resolution Protocol. RARP works opposite of ARP. Reverse Address Resolution Protocol is used to convert MAC address ie. physical address into IP address ie. logical address. RARP provides with a feature for the systems and applications to get their own IP address from a DNS(Domain Name System) or router. This type of resolution is required for various tasks such as executing reverse DNS lookup. As Reverse Address Resolution Protocol works at low level it requires direct network addresses. The reply from the server mostly carries a small information but the 32 bit internet address is used and it does not exploit the full potential of a network such as ethernet.

How Does RARP Work?

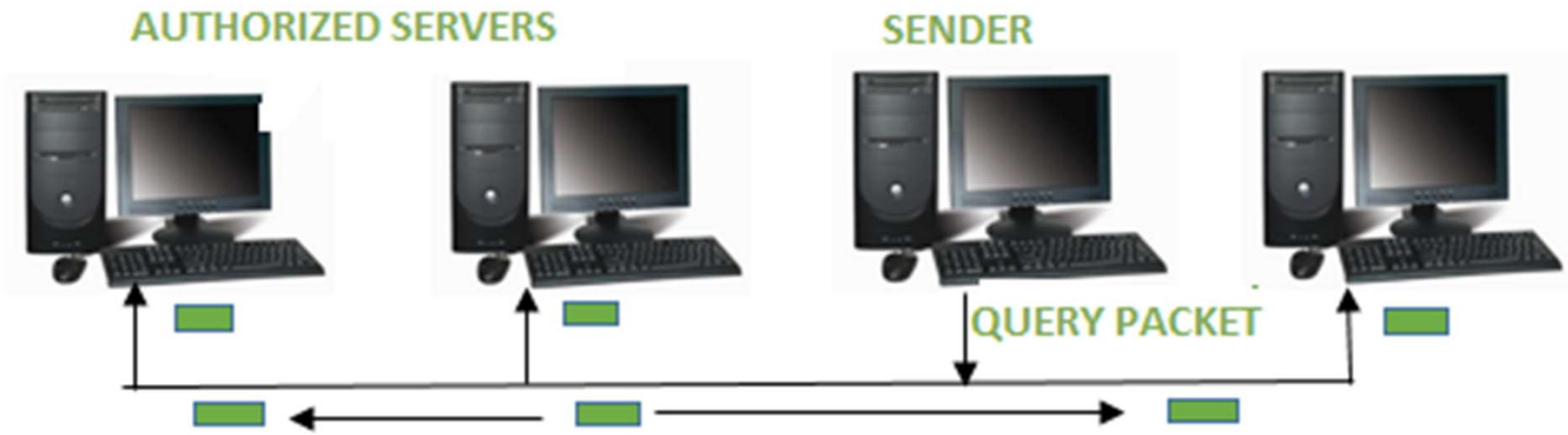
•Data is sent between two places in a network using the RARP, which is on the Network Access Layer.

•Every user on the network has two distinct addresses: their MAC (physical) address and their IP (logical) address.

•Software assigns the IP address, and the hardware then builds the MAC address into the device.

•Any regular computer connected to the network can function as the RARP server, answering to RARP queries. It must, however, store all of the MAC addresses' associated IP addresses. Only these RARP servers are able to respond to RARP requests that are received by the network. The information package must be transmitted over the network's lowest tiers.

•Using both its physical address and Ethernet broadcast address, the client transmits a RARP request. In response, the server gives the client its IP address.



REVERSE ADDRESS RESOLUTION PROTOCOL

ICMP

ICMP stands for Internet Control Message Protocol. ICMP is a part of IP protocol suite. ICMP is an error reporting and network diagnostic protocol. Feedback in the network is reported to the designated host. Meanwhile, if any kind of error occur it is then reported to ICMP. ICMP protocol consists of many error reporting and diagnostic messages. ICMP protocol handles various kinds of errors such as time exceeded, redirection, source quench, destination unreachable, parameter problems etc. The messages in ICMP are divided into two types. They are given below:

•**Error Message**: Error message states about the issues or problems that are faced by the host or routers during processing of IP packet.

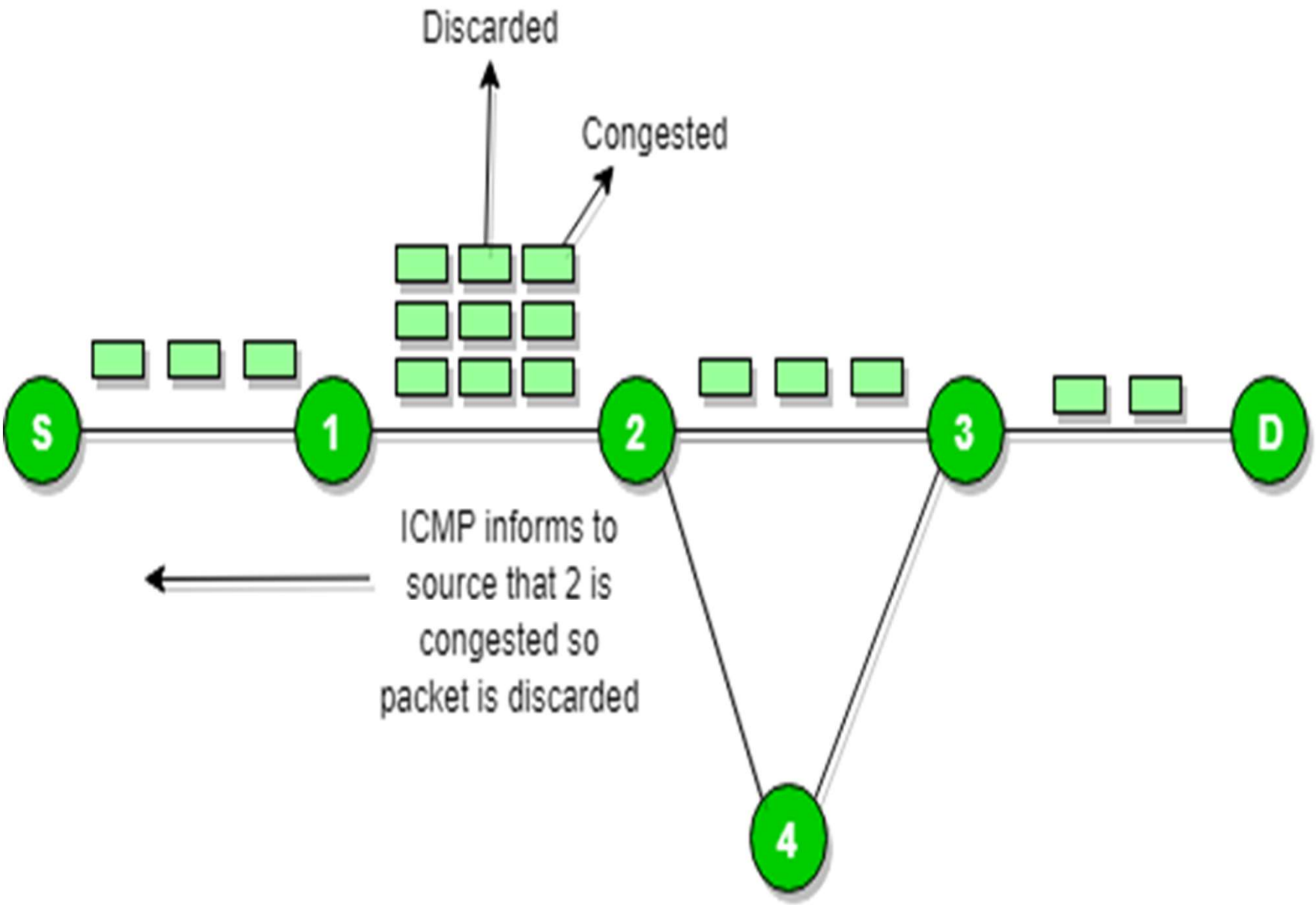
•**Query Message**: Query messages are used by the host in order to get information from a router or another host.

How Does ICMP Work?

•The main and most significant protocol in the IP suite is called ICMP. However, unlike TCP and UDP, ICMP is a connectionless protocol, meaning it doesn't require a connection to be established with the target device in order to transmit a message.

•TCP and ICMP operate differently from one another; TCP is a connection-oriented protocol, while ICMP operates without a connection. Every time a connection is made prior to a message being sent, a TCP Handshake is required of both devices.

•Datagrams including an IP header containing ICMP data are used to transmit ICMP packets. An independent data item like a packet is comparable to an ICMP datagram.


Discarded
Congested
S
1
2
3
D
4
ICMP informs to source that 2 is congested so packet is discarded

IGMP(Internet Group Message Protocol)

IGMP stands for Internet Group Message Protocol. IGMP is a multicasting communication protocol. It utilizes the resources efficiently while broadcasting the messages and data packets. IGMP is also a protocol used by TCP/IP. Other hosts connected in the network and routers makes use of IGMP for multicasting communication that have IP networks. In many networks multicast routers are used in order to transmit the messages to all the nodes. Multicast routers therefore receives large number of packets that needs to be sent. But to broadcast this packets is difficult as it would increase the overall network load. Therefore IGMP helps the multicast routers by addressing them while broadcasting. As multicast communication consists of more than one senders and receivers the Internet Group Message Protocol is majorly used in various applications such as streaming media, web conference tools, games, etc.

Routing protocols are the set of rules and algorithms that routers use to communicate with each other to find the most efficient path to transmit data packets from a sender to a receiver.

Unicast Routing Protocols

Unicast is a type of information transfer and it is used when there is a participation of single sender and single recipient. So, in short, you can term it as a one-to-one mapping. For example, a device having IP address 10.1.4.0 in a network wants to send the traffic stream (data packets) to the device with IP address 20.14.4.2 in the other network, then unicast comes into the picture. It is the most common form of data transfer over the networks.

Advantages of Unicast

•**Simplicity:** Unicast is very simple to implement as well as to manage since the data is transferred to a particular and explicit recipient.

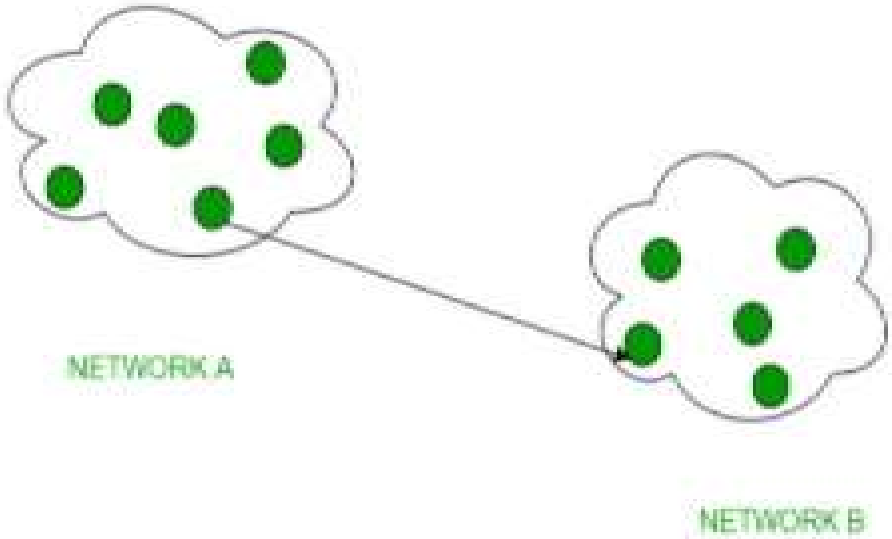
•**Reliable Communication:** As the data is sent to one receiver it provides the assurance of delivery and the acknowledgment from the receiver end which makes it ideal for reliable methods of communication such as TCP.

•**Security:** Since these are one-to-one communications, it is safer than the broadcast or multicast and can be used for transferring data.

Disadvantages of Unicast

•**Inefficiency for Group Communication:** For that reason, with the same data that needs to be sent to multiple receivers, Unicast may prove highly inefficient because each requires a separate transmission.

•**Increased Bandwidth Usage:** When data is sent to many receivers individually in large networks it leads to the wastage of bandwidth.



Multicast Routing Protocols

Multicasting has one/more senders and multiple recipients participate in data transfer traffic. In multicasting traffic recline between the boundaries of unicast and broadcast. It server's direct single copies of data streams and that are then simulated and routed to hosts that request it. IP multicast requires support of some other protocols such as Internet Group Management Protocol (IGMP), Multicast routing for its working. And also in Classful IP addressing Class D is reserved for multicast groups.

Advantages of Multicast

•**Efficiency:** Compared with unicast communication, multicast helps in cutting down the number of data that is transmitted across the network since the data transmitted can be received by several persons; this is otherwise known as 'Broadcast'.

•**Scalability:** Multicast is suitable for application that requires distribution of information or sharing resource to many users that can elicit the flood the network principle to narrow scale thus making it opportune for scalable applications.

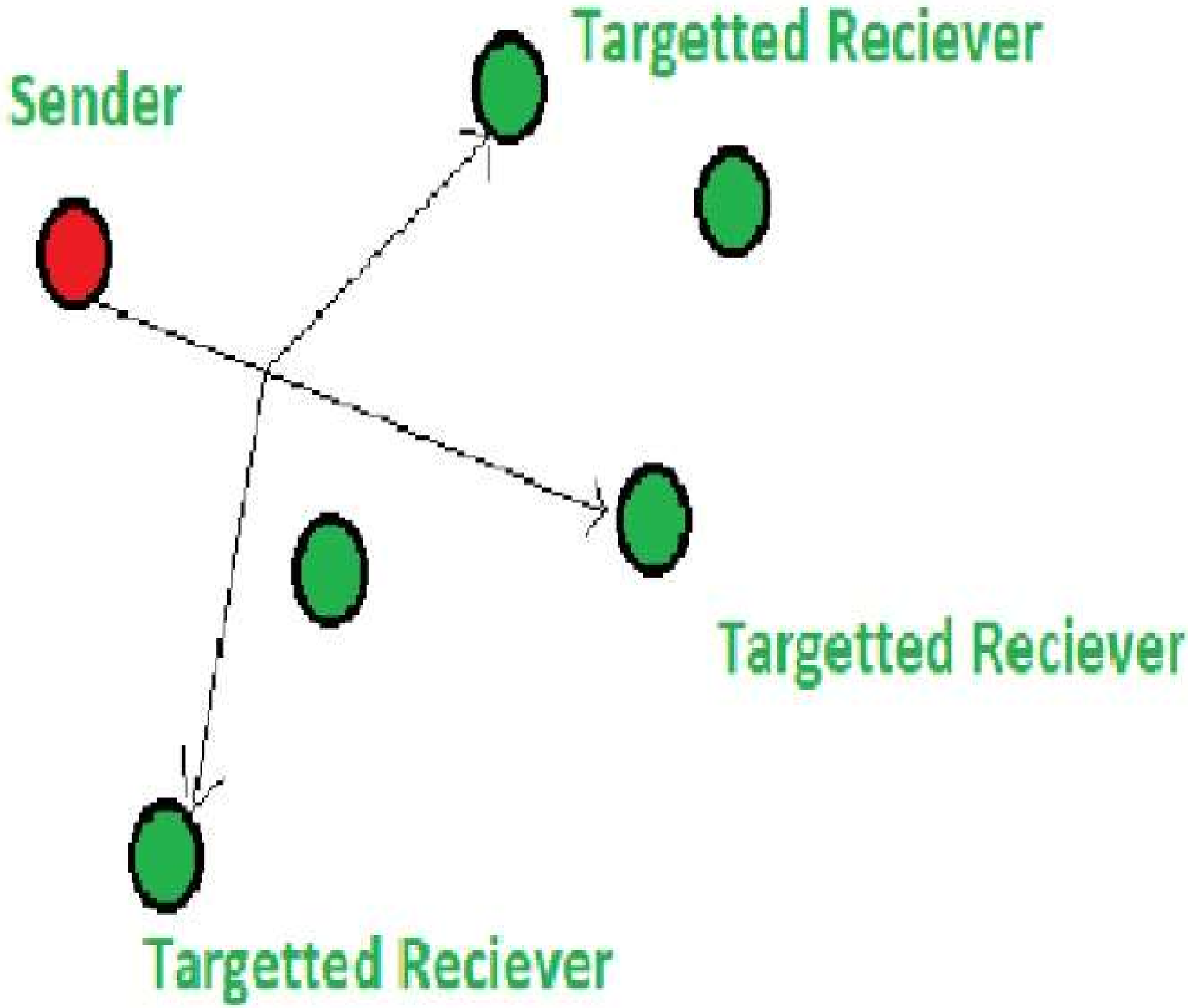
•**Cost-Effective:** Being a method that reduces the number of data set transmitted across the network, multicast helps in cutting down the overall expenditure on networks especially for the large scale broadcasts.

Disadvantages of Multicast

•**Complex Implementation:** Whereas in multicast setting, up and managing a multicast network is a little bit more complicated than unicast since an appropriate routing protocols and settings are necessary.

•**Less Reliable:** Multicast unfortunately can use connectionless protocols such as UDP unlike unicast thus making it incur in reliability as far as delivery is concerned.

•**Security Challenges:** Multicast communication is less secure as compared to unicast as the data is sent to multiple recipients thus posing a high risk of interception by unauthorized persons.

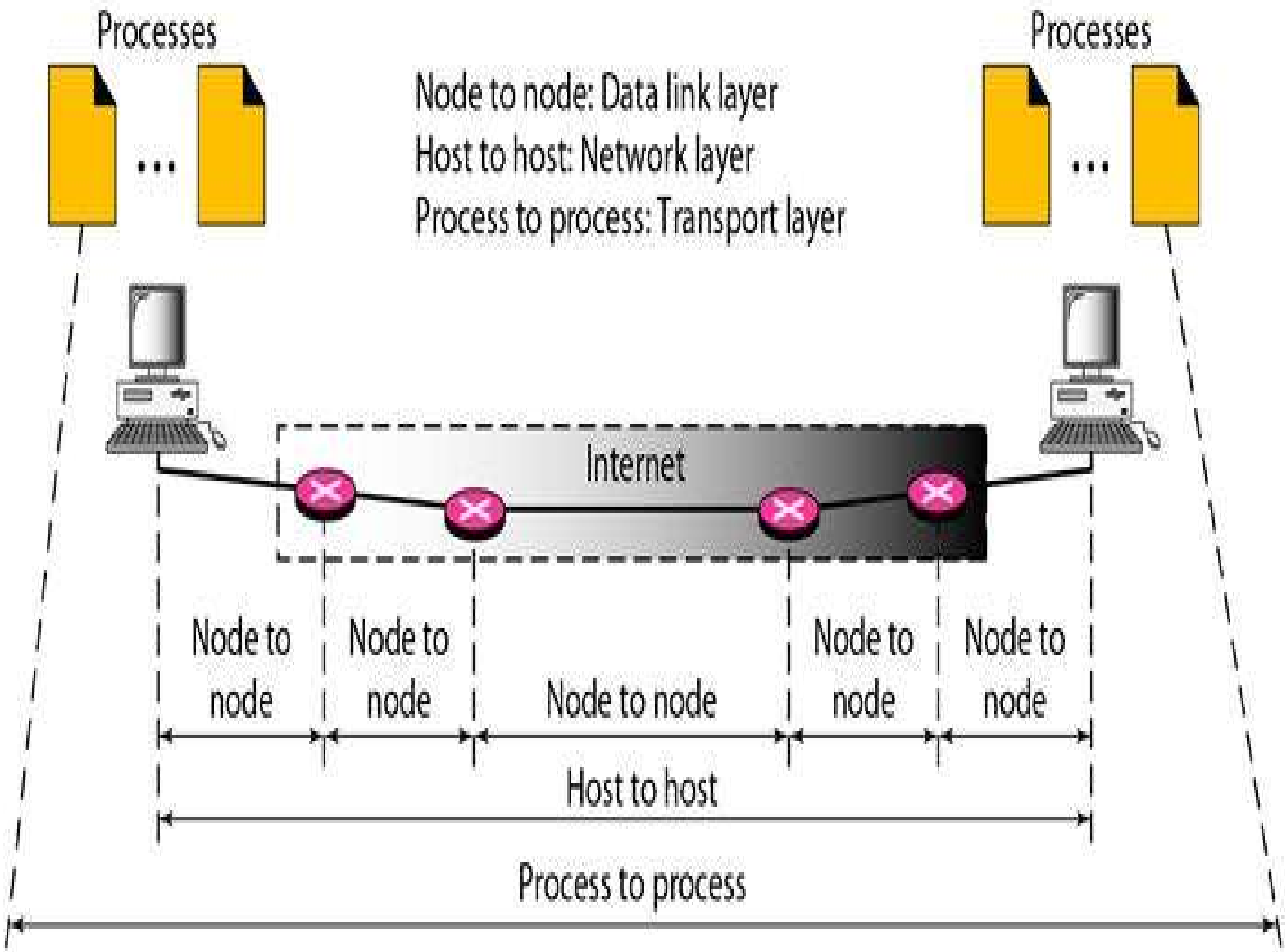

Targetted Reciever
Sender
Targetted Reciever
Targetted Reciever

Process to Process Delivery

The data link layer is responsible for delivery of frames between two neighboring nodes over a link. This is called node-to-node delivery. The network layer is responsible for delivery of datagrams between two hosts. This is called host-to-host delivery. Communication on the Internet is not defined as the exchange of data between two nodes or between two hosts. Real communication takes place between two processes. So that we need process-to-process delivery.

However, at any moment, several processes may be running on the source host and several on the destination host. To complete the delivery, we need a mechanism to deliver data from one of these processes running on the source host to the corresponding process running on the destination host.

The transport layer is responsible for process-to-process delivery-the delivery of a packet, part of a message, from one process to another. The following figure shows these three types of deliveries and their domains.


Processes
Processes
Node to node: Data link layer
Host to host: Network layer
Process to process: Transport layer
Internet
Node to node
Node to node
Node to node
Node to node
Node to node
Host to host
Process to process

The User Datagram Protocol (UDP)

The User Datagram Protocol (UDP) is simplest Transport Layer communication protocol available of the TCP/IP protocol suite. It involves minimum amount of communication mechanism. UDP is said to be an unreliable transport protocol but it uses IP services which provides best effort delivery mechanism.

In UDP, the receiver does not generate an acknowledgement of packet received and in turn, the sender does not wait for any acknowledgement of packet sent. This shortcoming makes this protocol unreliable as well as easier on processing.

TCP (Transmission Control Protocol)

TCP (Transmission Control Protocol) is one of the main protocols of the TCP/IP suite. It lies between the Application and Network Layers which are used in providing reliable delivery services. Transmission Control Protocol (TCP) ensures reliable and efficient data transmission over the internet. TCP plays a crucial role in managing the flow of data between computers, guaranteeing that information is delivered accurately and in the correct sequence.

The Domain Name System (DNS)

The Domain Name System (DNS) is like the internet's phone book. It helps you find websites by translating easy-to-remember names (like www.example.com) into the numerical IP addresses (like 192.0.2.1) that computers use to locate each other on the internet. Without DNS, you would have to remember long strings of numbers to visit your favorite websites.

Domain Name System (DNS) is a hostname used for IP address translation services. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers. It is required for the functioning of the Internet.

What is the Need for DNS?

Every host is identified by the IP address but remembering numbers is very difficult for people also the IP addresses are not static therefore a mapping is required to change the domain name to the IP address. So DNS is used to convert the domain name of the websites to their numerical IP address.

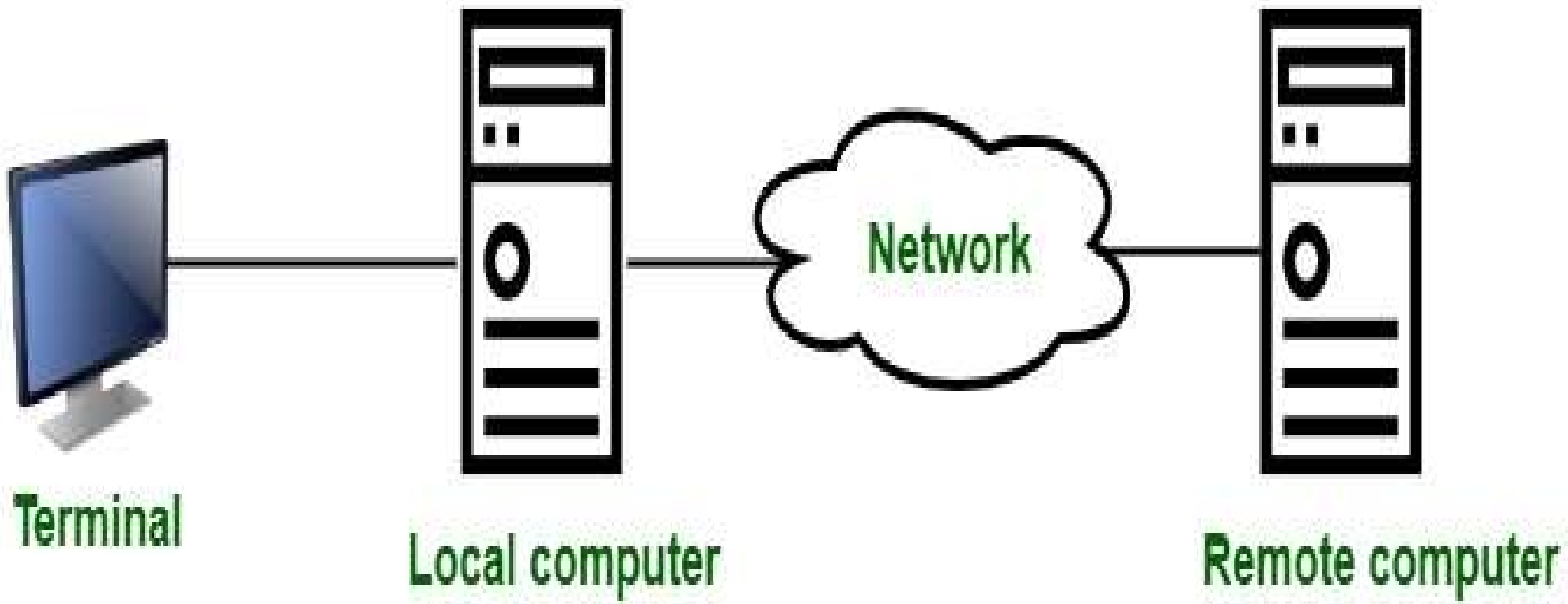
Dynamic DNS (DDNS)

Dynamic DNS (DDNS) is a service that can automatically update DNS records when an IP address changes. Many web properties, such as APIs or websites, run on internet connections that have their IP addresses changed frequently; this creates a problem if the operators of those properties want to give a hosted resource a specific domain name, which must then store an IP address in Domain Name System (DNS) records. Dynamic DNS (DDNS) is a service that keeps the DNS updated with a web property's correct IP address, even if that IP address is constantly being updated.

For example, if a web administrator is operating a small website with a domain name of www.example.com and an IP address of **192.0.2.0**, anytime another user enters www.example.com into their browser, the DNS will direct them to the server at **192.0.2.0**. If the admin's ISP dynamically changes the IP to **192.0.2.1**, a dynamic DNS service can automatically update the admin's DNS records so that other users trying to visit www.example.com will now go to the correct IP address.

Remote Login

Remote Login is a process in which user can login into remote site i.e. computer and use services that are available on the remote computer. With the help of remote login a user is able to understand result of transferring and result of processing from the remote computer to the local computer.



SMTP

An SMTP server's major function is to send, receive, and forward emails between the sender and the receiver. It does this by sending, receiving, and forwarding messages. The user's mail client or application can set the SMTP server's address, which is seen in the header of outgoing messages. The server address for Gmail is an example of a frequently seen type of SMTP server.

The simple mail transfer protocol (SMTP) server is not quite the same as a regular server, but an SMTP server does not necessarily need to be hosted on a machine. In order to send and receive e-mail, it functions more like an application that stays active all the time.

SMTP employs two techniques for sending and receiving email. End-to end and store-and-forward are two examples.

- End-to-end – The client's SMTP server will initiate the communication session throughout this transfer phase. The SMTP at the receiver's end, on the other hand, will respond to the client's request. Emails are sent to servers at multiple businesses using this form of transport. This model evolved as the favoured form of communication among various companies and domains.

- Store-and-forward – This mail transfer is only used when the sender and receiver are both from the same organization and the communication has been established using the receiver's SMTP. The SMTP server will make a copy of the mail and store it until the receiver's SMTP successfully receives the copy of the mail.

Advantages of Using SMTP

Following are some of the notable advantages of using SMTP –

- **Easy to Use** – It is one of the quickest methods for sending and receiving text messages over the internet. It is applicable to a wide range of systems, networks, and even tools. Using the internet, even the most basic computer can send an email over the world.
- **Long Reach** – You can send mail from anywhere in the world and it will be sent to the recipient in an instant. The SMTP protocol was created primarily to convey text messages across the internet. As a result, when you send an email, it receives a priority transfer.

•**Connect with Any System** – Emails have become an integral part of our daily lives. If you work in a firm, you understand how critical it is to monitor your emails. Emails may now be sent from any smartphone, eliminating the requirement for a computer or laptop nearby. Even a first-generation smartphone could send email. This demonstrates how adaptable the SMTP connection is. When it comes to compatibility, it truly has no bounds.

•**No Need of Development** – SMTP is already a well-established and widely used protocol. This transfer protocol requires no additional work on your part.

The Term "File Transfer Protocol" refers to the method that personal computers employ in order to transmit files over the Internet. A user who is the recipient of the files will be granted a unique set of rights, which may include the ability to open the file, make changes to it, send it back, or forward it. All of these permissions are set by the sender, who also has the ability to adjust them according to the needs of their organization.

FTP is an extremely important tool for web developers because it is used to keep the content of websites up to date. Transferring a significant quantity of files quickly is a requirement for website hosting. FTP is the method that is both the safest and the quickest when it comes to sending large files over the Internet.

For FTP to function properly, there must first be a setup in both the connection and the communicator. Users need to have authorization, which can be obtained by supplying credentials to the FTP server. Only then can they create these two parties.

Benefits of Using FTP

There are several methods available for sending files. However, some of the convincing arguments in favor of transferring data using FTP are as follows –

•Security – The provision of many layers of security is the primary advantage brought about by the utilization of an FTP server for the transmission of files. In addition, users have the ability to select the particular security mechanism that best suits their needs and use that for file transfers.

•Control – Another significant advantage is that you have complete command over the data transfer taking place. Even the smallest details of control have been handed over to you. In order to give you the ability to control who can upload files, download files, change files, and share files. FTP gives you the opportunity to operate your server in a more efficient manner, which means that not everyone in your company needs to have the same level of administrative power over the server.

•Bigger File Transfers – By using FTP servers, you won't have to

worry about the difficulty of delivering larger files to the intended recipient. You may transfer gigabytes of data in a matter of seconds with a dedicated FTP server if you configure it with a network connection capable of 10 gigabits per second.

Difference between FTP and SMTP

The following table highlights the major differences between FTP and SMTP –

Basis of comparison	FTP	SMTP
Full-Form	File Transfer Protocol	Simple Mail Transfer Protocol
Protocol	FTP's TCP connection is persistent for the control connection, but if TCP is used for data, the FTP protocol becomes non-persistent.	Transfer protocols mostly operate on the push protocol system.
Number of TCP Connections	2	1

Port number	FTP employs port numbers 20 and 21.	Port 25 is used by SMTP.
Type of Band	FTP makes use of out-of band transport techniques.	In-band transfer techniques are used for the SMTP connection.
Type of TCP Connection	Both Persistent and Non- Persistent	Persistent

The World Wide Web (WWW) is a networked information system that is made up of agents that exchange information. The architecture of the WWW is made up of three main components: identification, representation, and interaction:

- **Identification**

Agents use Uniform Resource Identifiers (URIs) to identify objects, or resources, in the system.

- **Representation**

Agents use a variety of data formats to describe and communicate the state of resources. These formats can be used separately or together, and include XHTML, CSS, PNG, XLink, RDF/XML, and SMIL animation.

- **Interaction**

Agents exchange representations using a variety of protocols, including HTTP, FTP, and SMTP.

Persistent HTTP	Non-Persistent HTTP
The server leaves the connection open after sending a response.	Requires 2 RTTs per object.
Subsequent HTTP messages between the same client/server are sent over an open connection.	OS overhead for each TCP connection
The client sends requests as soon as it encounters a referenced object.	Browsers often open parallel TCP connections to fetch referenced objects.
As little as one RTT for all the referenced objects.	Here, at most one object can be sent over one TCP Connection.

Wi-Fi(wireless fidelity) and Li-Fi(light fidelity) are two different technologies that are used to send and receive data wirelessly. In the case of Wi-Fi, we use Routers and Radio Frequency (RF) waves to transmit data, whereas in Li-Fi we use LED bulbs and Light signals to transmit and receive data.

What is WiFi?

Wireless fidelity, or wifi, is a wireless networking technique. It connects to networks and offers wireless internet using the IEEE 802.11 standard. Since cable requires installation and deployment on every computer to give an Internet connection, WiFi is more widely used because it offers a wireless connection.

Without the need for wires or cables, it functions similarly to a local area network. Wi-Fi is available on many modern computing products, including digital cameras, smart TVs, laptops, and cell phones.

Advantages of WiFi

•**Enhanced effectiveness**: Faster information transfer occurs between partners and customers as well as within firms when data connections are improved. While on a sales call, salespeople can, for instance, remotely verify prices and stock levels.

•**Savings on costs:** Installing wireless networks can be less expensive and easier, particularly in buildings that are listed or in situations where the landlord won't let wires be installed.

•**Fresh possibilities**: Using wireless networking might enable you to provide new goods or services. For instance, 'hot spot' WiFi services have been implemented in a lot of train stations, hotels, cafes, restaurants, and airport departure lounges to enable mobile users to connect their devices to their 'home' workplaces while on the road.

•**Availability and accessibility**: You are rarely disconnected thanks to wireless technology, which enables communication while on the go and eliminates the need for additional cables or adapters to access office networks.

•**Flexibility:** While out from the office, office-based wireless workers can still be productive because they can network without occupying a separate computer. This may give rise to new working practices including working from home or having direct access to company data while at a customer's location. Find out more about workers that do remote work.

Disadvantages of WiFi

•**Security:** It is especially important to secure your wireless systems since unauthorised people are more likely to assault wireless transmissions.

•**Installation issues**: If there are additional sources of electromagnetic (radio) interference or if other people in the same building utilise wireless equipment, you can experience interference. In severe circumstances, this can result in a total loss of wireless connectivity or inadequate communication.

•**Reportage:** It might be challenging to provide reliable coverage in some buildings, which can result in "black spots" where the signal is unavailable. For instance, it could be challenging for you to hear the radio frequencies utilised in buildings constructed with steel reinforcing components.

•**Transfer rates:** Compared to 'wired' networks, wireless transmission can be less effective and slower.

What is LiFi?

Lifi, which stands for Light Fidelity, transfers and receives data via light signals and LED bulbs. Light has been used as an illumination source in homes, offices, and other spaces for a number of decades. But in the modern day, LEDs are utilised for communication in addition to lighting. LEDs are now capable of serving as a wireless communication link to access the World Wide Web from a computer or smart device.

Advantages of LiFi

•**Bandwidth and Speed**: LiFi is able to provide mobile devices with several Gbps speeds. By pushing wireless technology beyond its present limits, this next generation technology will create previously unheard-of bandwidth.

•**Security of a Military Kind**: Within a physical place, light can be secured and contained. Since LiFi provides accurate localization for asset monitoring and user authentication, it allows for much more control.

•**No Congestion**: Unlike WiFi and other RF technologies, LiFi is not affected by congestion or the negative consequences of multiuser degradation.

•**Reliable**: LiFi offers improved stability that makes it possible to communicate without interference and 1000 times more data density, which significantly enhances user experience.

•**Minimal Latency:** Presently, LiFi provides three times less latency than Wi-Fi and has the potential to significantly boost automation, creativity, and applications like augmented and virtual reality.

Disadvantages of LiFi

•Only in areas with available source light may one use the internet. Furthermore, light can only travel in a straight line of sight path and cannot pass through walls. This restricts internet access wherever it is needed. Its range is also constrained.

•It cannot be used in an outdoor setting where RF signals are present. This is a result of interference from surrounding optical sources and sunlight. Furthermore, if it is used outside, it may be intercepted by unauthorised individuals.

•Even though it is easy to install, a Lifi system needs a completely new infrastructure. This will increase the cost for businesses or individuals wishing to use LiFi Internet service.

•Lights must be kept on day and night in order to use wifi internet services, even if it uses little power.